A Unified Framework for Adsorption Isotherm Analysis in Water Quality Studies: Descriptors, Model Selection, and Reporting Guidelines

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How to read this SI

This SI is written as a practical route from raw isotherm data to reportable descriptors. For each model, ask three questions: **(1)** does it have a real saturation capacity $Q_{\max}$? **(2)** does it provide a comparable affinity K^*_{aff}? **(3)** what heterogeneity descriptor should be reported?

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Nomenclature used for reporting

The final reporting descriptors are

$$\{Q_{\max},\, K^*_{\text{aff}},\, E_{\text{ads}},\, \sigma_H\}.$$

Here $\Delta G^0_{\text{ads}} = -RT \ln K^*_{\text{aff}}$ is the standard Gibbs free energy associated with the declared standard state. The reported positive affinity-energy descriptor is $E_{\text{ads}} \equiv E_{\text{aff}} = -\Delta G^0_{\text{ads}} = RT \ln K^*_{\text{aff}}$. Larger E_{ads} therefore means stronger standard-state affinity. The standard-state concentration may appear as C° or C^0; both mean the declared concentration standard used to make equilibrium constants dimensionless. The symbol σ_E denotes the apparent mathematical width of an energy distribution; σ_H is the reported intrinsic heterogeneity width.

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Warning

Do **not** report $Q_{\max}$ from the Freundlich equation. The Freundlich model has no saturation plateau; any capacity derived from it depends on the chosen concentration window and is physically meaningless. FREUNDLICH

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S-1 Isotherm model equations and limits

This section collects the full equations and limiting behaviors used in Section 2 of the main manuscript. The availability summary is given in Table 1 of the main manuscript.

Reader rule

For every isotherm, separate the fitted parameters from the reported descriptors. Fitted parameters belong to one mathematical model; descriptors are the quantities used to compare materials across different models.

S-1.1 Henry isotherm

The Henry isotherm is the low-coverage limit of many models:

$$q_e = K_H C_e. \tag{S-1}$$

It has no saturation capacity $Q_{\max}$ and therefore no model-harmonized $K_{\text{aff}} = K_H/Q_{\max}$. The Henry constant K_H is defined by the initial slope.

Low-concentration limit. The Henry model is itself the $C_e \to 0$ limit:

$$\lim_{C_e \to 0} \frac{q_e}{C_e} = K_H. \tag{S-2}$$

High-concentration limit. The model is linear and **does not saturate**, so $q_e \to \infty$ as $C_e \to \infty$.

Take-home message

Henry's law is useful for the initial slope, but it is not a full capacity model. Use K_H to describe dilute uptake; do not use the Henry equation to report $Q_{\max}$.

S-1.2 Langmuir isotherm

The Langmuir isotherm is written as

$$q_e = \frac{Q_{\max} K_L C_e}{1 + K_L C_e}, \tag{S-3}$$

where q_e is the equilibrium adsorbed amount, C_e is the equilibrium concentration in the fluid phase, $Q_{\max}$ is the maximum adsorption capacity, and K_L is the Langmuir equilibrium constant.

S-1.2.1 High concentration limit, $Q_{\max}$: $C_e \to \infty$

From Eq. (S-3),

$$\lim_{C_e\to\infty} q_e = \lim_{C_e\to\infty} \frac{Q_{\max} K_L C_e}{1 + K_L C_e} = Q_{\max}. \tag{S-4}$$

Therefore,

Capacity-limit result

Langmuir has a finite saturation capacity:

$$\lim_{C_e\to\infty} q_e = Q_{\max}. \tag{S-5}$$

This makes Langmuir the clean reference model for reporting $Q_{\max}$ and for defining the homogeneous limit $\sigma_H = 0$.

S-1.2.2 Low concentration limit: Henry constant

For small C_e with $K_L C_e \ll 1$,

$$q_e \approx Q_{\max} K_L C_e. \tag{S-6}$$

The Henry law form is

$$q_e \approx K_H C_e, \tag{S-7}$$

so the Henry constant is the initial slope:

$$K_H = \left.\frac{\mathrm{d}q_e}{\mathrm{d}C_e}\right|_{C_e \to 0} = Q_{\max} K_L. \tag{S-8}$$

Thus,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = Q_{\max} K_L. \tag{S-9}$$

S-1.2.3 Adsorption free energy

Introduce a standard concentration C^0 and define

$$K_L^* = K_L C^0. \tag{S-10}$$

The standard Gibbs free energy of adsorption is

$$\Delta G^0_{\mathrm{ads}} = -RT \ln\left(K_L C^0\right), \tag{S-11}$$

and the corresponding affinity energy is

$$E_{\mathrm{aff}} = -\Delta G^0_{\mathrm{ads}} = RT \ln\left(K_L C^0\right). \tag{S-12}$$

Equivalently,

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$\Delta G^0_{\text{ads}} = -RT \ln\left(K_L C^0\right), \qquad E_{\text{aff}} = RT \ln\left(K_L C^0\right). \tag{S-13}$$

Take-home message

Langmuir is the reference homogeneous model. It gives a finite $Q_{\max}$, a finite Henry constant, and $\sigma_H = 0$. Any heterogeneous model that becomes Langmuir in a limiting case should reduce smoothly to these three values.

S-1.3 Double Langmuir isotherm

The dual-site (double) Langmuir isotherm is

$$q_e = \frac{Q_{\max,1} K_1 C_e}{1 + K_1 C_e} + \frac{Q_{\max,2} K_2 C_e}{1 + K_2 C_e}, \tag{S-14}$$

where $Q_{\max,i}$ and K_i are the site capacities and constants.

S-1.3.1 High concentration limit: $C_e \to \infty$

For each site,

$$\lim_{C_e \to \infty} \frac{Q_{\max,i} K_i C_e}{1 + K_i C_e} = Q_{\max,i}, \tag{S-15}$$

so the total saturation capacity is finite.

Capacity-limit result

Double Langmuir has a finite total saturation capacity:

$$\lim_{C_e \to \infty} q_e = Q_{\max,1} + Q_{\max,2}. \tag{S-16}$$

Report the sum as the total capacity, but keep $Q_{\max,1}$, $Q_{\max,2}$, K_1, and K_2 when the goal is to interpret two distinct site families.

S-1.3.2 Low concentration limit: global Henry constant

For $K_i C_e \ll 1$,

$$q_e \approx Q_{\max,1} K_1 C_e + Q_{\max,2} K_2 C_e, \tag{S-17}$$

which has the Henry form (Eq. (S-7)) with

$$K_H^{(\text{tot})} = Q_{\max,1} K_1 + Q_{\max,2} K_2. \tag{S-18}$$

S-1.3.3 Effective Langmuir constant

Define $Q_{\max}^{(\text{tot})} = Q_{\max,1} + Q_{\max,2}$. Then

$$K_L^{(\text{tot})} = \frac{K_H^{(\text{tot})}}{Q_{\max}^{(\text{tot})}} = \frac{Q_{\max,1} K_1 + Q_{\max,2} K_2}{Q_{\max,1} + Q_{\max,2}}, \tag{S-19}$$

which is a capacity-weighted average of K_1 and K_2.

S-1.3.4 Thermodynamic interpretation

With a standard concentration C^0,

$$K_i^* = K_i C^0, \qquad K_L^{(\text{tot})*} = K_L^{(\text{tot})} C^0. \tag{S-20}$$

The effective adsorption free energy and affinity are

$$\Delta G^{0,(\mathrm{tot})}_{\mathrm{ads}} = -RT\ln\left(K_L^{(\mathrm{tot})}C^0\right), \qquad E^{(\mathrm{tot})}_{\mathrm{aff}} = RT\ln\left(K_L^{(\mathrm{tot})}C^0\right). \tag{S-21}$$

S-1.4 Freundlich isotherm

The Freundlich isotherm is

$$q_e = K_F C_e^{1/n}, \tag{S-22}$$

where K_F is a capacity-related constant and $1/n$ is the heterogeneity exponent.

S-1.4.1 High concentration limit: no intrinsic $Q_{\max}$

Taking $C_e \to \infty$ in Eq. (S-22),

$$\lim_{C_e\to\infty} q_e = \infty. \tag{S-23}$$

Thus, the Freundlich model has no finite saturation capacity:

Warning

$$\text{Freundlich has no intrinsic } Q_{\max}. \tag{S-24}$$

S-1.4.2 Low concentration behavior and Henry law

The derivative of Eq. (S-22) is

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = K_F \frac{1}{n} C_e^{1/n-1}. \tag{S-25}$$

For $1/n = 1$, the Freundlich model reduces to Henry law with $K_H = K_F$. For $1/n \neq 1$, the initial slope diverges ($1/n < 1$) or vanishes ($1/n > 1$) as $C_e \to 0$. An apparent, concentration-dependent Henry coefficient can be defined as

$$K_H(C_e) = \frac{q_e}{C_e} = K_F C_e^{1/n-1}. \tag{S-26}$$

S-1.4.3 Affinity constant and apparent free energy

Using an effective Henry coefficient at a reference concentration C_{ref}, one can define

$$K^*_{\text{eff}} = \left[\frac{1}{n} K_F C_{\text{ref}}^{1/n-1}\right] C^0, \tag{S-27}$$

leading to an apparent free energy

$$\Delta G^{0,\text{app}}_{\text{ads}}(C_{\text{ref}}) = -RT \ln\left[\frac{1}{n} K_F C_{\text{ref}}^{1/n-1} C^0\right]. \tag{S-28}$$

This dependence on C_{ref} reflects the lack of a unique free energy for the Freundlich model.

Reporting warning

Freundlich can fit curved data, but it does not define a unique Q_{max}, K^*_{aff}, or E_{ads} without an arbitrary concentration reference. Treat Freundlich affinity values as local or apparent, not as universal descriptors.

S-1.5 Sips (Langmuir–Freundlich) isotherm

The Sips model introduces a finite saturation capacity while retaining heterogeneity:

$$q_e = \frac{Q_{\text{max}} K_S C_e^m}{1 + K_S C_e^m}, \tag{S-29}$$

where $0 < m \leq 1$. When $m = 1$, the model reduces to Langmuir.

S-1.5.1 High concentration limit: finite Q_{max}

Taking $C_e \to \infty$ in Eq. (S-29),

$$\lim_{C_e \to \infty} q_e = Q_{\text{max}}. \tag{S-30}$$

S-1.5.2 Low concentration limit: Freundlich behavior

For $K_S C_e^m \ll 1$,

$$q_e \approx Q_{\max} K_S C_e^m. \tag{S-31}$$

This has the Freundlich form with

$$K_F = Q_{\max} K_S, \qquad \frac{1}{n} = m. \tag{S-32}$$

S-1.5.3 Initial slope and local Henry coefficient

The low-concentration ratio is

$$\frac{q_e}{C_e} \approx Q_{\max} K_S C_e^{m-1}, \tag{S-33}$$

so a local Henry coefficient can be defined as

$$K_H(C_e) = Q_{\max} K_S C_e^{m-1}, \tag{S-34}$$

which is constant only when $m = 1$.

S-1.5.4 Effective affinity energy

Define an effective equilibrium constant K_{eq} by

$$\frac{E_0}{RT} = \ln\bigl(K_{\mathrm{eq}}\, C^\circ\bigr). \tag{S-35}$$

For Langmuir, $K_{\mathrm{eq}} = K_L$, so

$$\text{Langmuir:} \qquad \frac{E_0}{RT} = \ln\bigl(K_L C^\circ\bigr). \tag{S-36}$$

For Sips, the constant K_S has units $[C]^{-m}$, so an effective equilibrium constant with units $[C]^{-1}$ is

$$K_{\text{eq}}^{(\text{Sips})} = K_S^{1/m}. \tag{S-37}$$

Therefore,

$$\frac{E_0}{RT} = \frac{1}{m} \ln\big(K_S\,(C^\circ)^m\big). \tag{S-38}$$

In the limit $m = 1$, Eq. (S-38) reduces to Eq. (S-36).

Sips convention

This work uses $q_e = Q_{\max} K_S C_e^m/(1 + K_S C_e^m)$. Therefore K_S has units $[C]^{-m}$, and the comparable affinity is $K_S^{1/m}$, which has units $[C]^{-1}$. This is the quantity used in K^*_{aff}.

S-1.6 Tóth isotherm

The Tóth isotherm is

$$q_e = \frac{q_{\max}\, b\, C_e}{\left[1 + (bC_e)^t\right]^{1/t}}, \tag{S-39}$$

where $q_{\max}$ is the saturation capacity, b is the affinity constant, and $0 < t \leq 1$ is the heterogeneity parameter. When $t = 1$, the model reduces to Langmuir.

S-1.6.1 High concentration limit: finite $Q_{\max}$

As $C_e \to \infty$, $(bC_e)^t \gg 1$, so

$$\lim_{C_e\to\infty} q_e = \lim_{C_e\to\infty} \frac{q_{\max}\, b\, C_e}{(bC_e)^{t/t}} = q_{\max}. \tag{S-40}$$

Capacity-limit result

Tóth has a finite saturation capacity:

$$\lim_{C_e \to \infty} q_e = q_{\max}. \tag{S-41}$$

This $q_{\max}$ can be reported as $Q_{\max}$, while the shape parameter t controls how strongly the curve departs from Langmuir.

S-1.6.2 Low concentration limit: Henry constant

For $bC_e \ll 1$, $(bC_e)^t \ll 1$ and $[1 + (bC_e)^t]^{1/t} \approx 1$, so

$$q_e \approx q_{\max}\, b\, C_e. \tag{S-42}$$

Note

$$K_H = q_{\max}\, b. \tag{S-43}$$

S-1.6.3 Affinity constant and adsorption free energy

The affinity constant is $K_{\text{aff}} = b$ (units $[C]^{-1}$). With a standard concentration C^0,

$$K^*_{\text{aff}} = b\, C^0, \qquad E_{\text{aff}} = RT \ln(b\, C^0). \tag{S-44}$$

Note

$$\begin{aligned} \Delta G^0_{\text{ads}} &= -RT \ln\left(b\, C^0\right), \\ E_{\text{aff}} &= RT \ln\left(b\, C^0\right). \end{aligned} \tag{S-45}$$

S-1.7 Jovanovic isotherm

The Jovanovic isotherm is

$$q_e = q_{\max}\left[1 - \exp(-b\,C_e)\right]. \tag{S-46}$$

S-1.7.1 High concentration limit: finite $Q_{\max}$

As $C_e \to \infty$, $\exp(-bC_e) \to 0$, so

$$\lim_{C_e\to\infty} q_e = q_{\max}. \tag{S-47}$$

Capacity-limit result

Jovanovic has a finite saturation capacity:

$$\lim_{C_e\to\infty} q_e = q_{\max}. \tag{S-48}$$

For reporting, $q_{\max}$ plays the same capacity role as $Q_{\max}$ in Langmuir-type models.

S-1.7.2 Low concentration limit: Henry constant

For $bC_e \ll 1$, $\exp(-bC_e) \approx 1 - bC_e$, so

$$q_e \approx q_{\max}\,b\,C_e. \tag{S-49}$$

Therefore,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = q_{\max}\,b. \tag{S-50}$$

S-1.7.3 Affinity constant and adsorption free energy

The affinity constant is $K_{\text{aff}} = b$. With a standard concentration C^0,

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$K^*_{\text{aff}} = b\,C^0, \qquad E_{\text{aff}} = RT\ln(b\,C^0). \tag{S-51}$$

S-1.8 Temkin isotherm

The Temkin isotherm is

$$q_e = \frac{RT}{b_T}\ln(a_T\,C_e). \tag{S-52}$$

S-1.8.1 High concentration limit: no intrinsic $Q_{\max}$

As $C_e \to \infty$, $\ln(a_T C_e) \to \infty$, so

$$\lim_{C_e\to\infty} q_e = \infty. \tag{S-53}$$

Warning

$$\text{Temkin has no intrinsic } Q_{\max}. \tag{S-54}$$

The Temkin model is valid only over an intermediate concentration range where the linear-in-$\ln C_e$ approximation holds.

S-1.8.2 Low concentration limit: no finite Henry constant

The derivative is

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = \frac{RT}{b_T\,C_e}, \tag{S-55}$$

which diverges as $C_e \to 0$. Therefore,

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$\text{Temkin has no finite } K_H. \tag{S-56}$$

S-1.8.3 Effective affinity constant and adsorption free energy

An effective affinity constant is defined from the binding constant a_T (units $[C]^{-1}$):

$$K_{\text{aff}} = a_T, \qquad K^*_{\text{aff}} = a_T\, C^0. \tag{S-57}$$

The adsorption free energy is

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$E_{\text{aff}} = RT \ln(a_T\, C^0). \tag{S-58}$$

Note that this is an *effective* energy; the Temkin model describes the mid-coverage regime and does not correspond to a well-defined thermodynamic equilibrium constant.

S-1.9 UNILAN isotherm

The UNILAN isotherm (uniform-energy Langmuir) is

$$q_e = \frac{Q_{\max}}{2s} \ln\left(\frac{1 + b_{\max}\, C_e}{1 + b_{\min}\, C_e}\right), \tag{S-59}$$

where $s = \frac{1}{2} \ln(b_{\max}/b_{\min})$ is the half-width of the uniform energy distribution.

S-1.9.1 High concentration limit: finite $Q_{\max}$

As $C_e \to \infty$, $b_{\max} C_e \gg 1$ and $b_{\min} C_e \gg 1$, so

$$q_e \approx \frac{Q_{\max}}{2s} \ln\left(\frac{b_{\max}}{b_{\min}}\right) = \frac{Q_{\max}}{2s} \cdot 2s = Q_{\max}. \tag{S-60}$$

Capacity-limit result

UNILAN has a finite saturation capacity:

$$\lim_{C_e \to \infty} q_e = Q_{\max}. \tag{S-61}$$

Here $Q_{\max}$ is retained as the capacity descriptor, while the energy-band width is carried separately by $b_{\max}/b_{\min}$.

S-1.9.2 Low concentration limit: Henry constant

For $b_{\max} C_e \ll 1$ and $b_{\min} C_e \ll 1$, $\ln(1 + x) \approx x$, so

$$q_e \approx \frac{Q_{\max}}{2s} (b_{\max} - b_{\min})\, C_e. \tag{S-62}$$

Therefore,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = \frac{Q_{\max}}{2s} (b_{\max} - b_{\min}). \tag{S-63}$$

S-1.9.3 Affinity constant and adsorption free energy

The natural affinity constant is the geometric mean of the site constants:

$$K_{\text{aff}} = \sqrt{b_{\max}\, b_{\min}}, \qquad K^{*}_{\text{aff}} = \sqrt{b_{\max}\, b_{\min}}\; C^{0}. \tag{S-64}$$

The adsorption free energy is

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$E_{\text{aff}} = RT \ln\!\Big(\sqrt{b_{\max}\, b_{\min}}\; C^{0}\Big) = \frac{RT}{2} \ln\!\big(b_{\max}\, b_{\min}\,(C^{0})^{2}\big)\,. \tag{S-65}$$

S-1.10 Redlich–Peterson isotherm

The Redlich–Peterson isotherm is

$$q_e = \frac{K_{RP}\, C_e}{1 + a_{RP}\, C_e^{\beta}}, \tag{S-66}$$

with $0 < \beta \le 1$.

S-1.10.1 High concentration limit: conditional $Q_{\max}$

As $C_e \to \infty$,

$$q_e \approx \frac{K_{RP}}{a_{RP}}\, C_e^{1-\beta}. \tag{S-67}$$

If $\beta = 1$ (Langmuir limit): $Q_{\max} = K_{RP}/a_{RP}$ (finite). If $\beta < 1$: $q_e \to \infty$ (no finite saturation).

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$Q_{\max} = K_{RP}/a_{RP} \quad \text{(only if } \beta = 1\text{)}. \tag{S-68}$$

S-1.10.2 Low concentration limit: Henry constant

For $a_{RP}C_e^{\beta} \ll 1$,

$$q_e \approx K_{RP}\, C_e. \tag{S-69}$$

Therefore, regardless of β,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = K_{RP}. \tag{S-70}$$

S-1.10.3 Affinity constant and adsorption free energy

Two effective affinity constants are commonly used:

- Henry regime: $K_{\text{aff}} = K_{RP}/Q_{\max} = a_{RP}$ (when $\beta = 1$).
- High-C regime: $K_{\text{aff}} = (a_{RP})^{1/\beta}$.

For the purpose of cross-model comparison, we adopt the Henry-based definition:

Capacity-limit result

This is the saturation capacity that can be reported as a finite capacity descriptor. If the equation includes a condition, the descriptor is valid only under that condition.

$$K^{*}_{\mathrm{aff}} = \frac{K_{RP}}{Q_{\max}}\, C^0 = a_{RP}\, C^0 \quad (\beta = 1), \qquad E_{\mathrm{aff}} = RT \ln(a_{RP}\, C^0). \tag{S-71}$$

S-1.11 Koble–Corrigan isotherm

The Koble–Corrigan isotherm is

$$q_e = \frac{A_{KC}\, C_e^{1/n}}{1 + B_{KC}\, C_e^{1/n}}. \tag{S-72}$$

This is algebraically equivalent to the Sips isotherm with the identifications $Q_{\max} = A_{KC}/B_{KC}$, $K_S = B_{KC}$, and $m = 1/n$.

S-1.11.1 High concentration limit: finite $Q_{\max}$

As $C_e \to \infty$, $B_{KC} C_e^{1/n} \gg 1$, so

$$\lim_{C_e \to \infty} q_e = \frac{A_{KC}}{B_{KC}}. \tag{S-73}$$

Capacity-limit result

This is the saturation capacity that can be reported as a finite capacity descriptor. If the equation includes a condition, the descriptor is valid only under that condition.

$$Q_{\max} = A_{KC}/B_{KC}. \tag{S-74}$$

S-1.11.2 Low concentration limit: conditional Henry constant

For $B_{KC} C_e^{1/n} \ll 1$,

$$q_e \approx A_{KC}\, C_e^{1/n}. \tag{S-75}$$

A finite K_H exists only when $1/n = 1$: $K_H = A_{KC}$. For $1/n \neq 1$, the initial slope is concentration dependent.

S-1.11.3 Affinity constant and adsorption free energy

By analogy with Sips, the effective affinity constant is $K_{\text{aff}} = B_{KC}^n$, so

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$K^*_{\text{aff}} = B_{KC}^n\,C^0, \qquad E_{\text{aff}} = nRT\ln(B_{KC}\,C^0). \tag{S-76}$$

S-1.12 Radke–Prausnitz isotherm

The Radke–Prausnitz isotherm is

$$q_e = \frac{a_{RP}\,r_{RP}\,C_e}{a_{RP} + r_{RP}\,C_e^{1-p}}, \tag{S-77}$$

with $0 < p \leq 1$.

S-1.12.1 High concentration limit: conditional $Q_{\max}$

As $C_e \to \infty$, for $p < 1$:

$$q_e \approx \frac{a_{RP}\,r_{RP}}{r_{RP}}\,C_e^p = a_{RP}\,C_e^p \to \infty. \tag{S-78}$$

For $p = 1$: $q_e \to a_{RP}\,r_{RP}/(a_{RP} + r_{RP}) \cdot C_e$, which also diverges.

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$\text{Radke–Prausnitz has no intrinsic } Q_{\max} \text{ for } p < 1. \tag{S-79}$$

S-1.12.2 Low concentration limit: Henry constant

For small C_e, $r_{RP} C_e^{1-p} \ll a_{RP}$, so

$$q_e \approx r_{RP}\, C_e. \tag{S-80}$$

Therefore,

Henry-limit result

This is the low-concentration slope. Use it to connect the fitted isotherm to the Henry regime, and compare it across models only when the concentration units and standard state are declared.

$$K_H = r_{RP}. \tag{S-81}$$

S-1.12.3 Effective affinity constant and adsorption free energy

Since no unique $Q_{\max}$ exists, the affinity constant is defined directly from the Henry constant:

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$K^*_{\text{aff}} = r_{RP}\, C^0, \qquad E_{\text{aff}} = RT \ln(r_{RP}\, C^0). \tag{S-82}$$

This is an *effective* energy valid in the low-concentration regime.

S-1.13 Dubinin–Radushkevich (D-R) and Dubinin–Astakhov (D-A)

The D-R/D-A isotherms are

$$q_e = Q_{\max} \exp\left[-\left(\frac{\varepsilon}{E_a}\right)^{n_{DA}}\right], \qquad \varepsilon = RT \ln\left(1 + \frac{C_{\mathrm{ref}}}{C_e}\right), \tag{S-83}$$

The D-R model corresponds to $n_{DA} = 2$. Here, C_{ref} is a reference concentration (typically $1\ \mathrm{mg\,L^{-1}}$ or $1\ \mathrm{mol\,L^{-1}}$) introduced to maintain dimensional consistency, as adding 1 to a reciprocal concentration with physical units is mathematically invalid.

Normalization convention

The Polanyi potential is written with an explicit reference concentration, $\varepsilon = RT\ln(1 + C_{\mathrm{ref}}/C_e)$, so the logarithm remains dimensionless. The chosen C_{ref} must be reported in the same concentration units as C_e.

S-1.13.1 High concentration limit: finite $Q_{\max}$

As $C_e \to \infty$, $\varepsilon = RT\ln(1 + C_{\mathrm{ref}}/C_e) \to 0$, so $\exp[-(\varepsilon/E_a)^{n_{DA}}] \to 1$ and

$$\lim_{C_e\to\infty} q_e = Q_{\max}. \tag{S-84}$$

Capacity-limit result

D-R/D-A has a finite saturation capacity:

$$\lim_{C_e\to\infty} q_e = Q_{\max}. \tag{S-85}$$

The capacity is well defined, but the affinity scale must still be normalised through the declared C_{ref} and C^0.

S-1.13.2 Low concentration limit: no finite Henry constant

As $C_e \to 0$, $\varepsilon \to \infty$, so $\exp[-(\varepsilon/E_a)^{n_{DA}}] \to 0$ faster than any power of C_e. The derivative

$$\frac{\mathrm{d}q_e}{\mathrm{d}C_e} = Q_{\max} \frac{n_{DA}}{E_a} \left(\frac{\varepsilon}{E_a}\right)^{n_{DA}-1} \exp\left[-\left(\frac{\varepsilon}{E_a}\right)^{n_{DA}}\right] \cdot \frac{RT\, C_{\text{ref}}}{C_e(C_e + C_{\text{ref}})} \tag{S-86}$$

diverges as $C_e \to 0$.

Reporting limitation

This is a limitation, not a descriptor to force into the reporting table. State it explicitly so the reader does not compare a missing capacity, Henry constant, or affinity as though it were directly defined by the model.

$$\text{D-R/D-A has no finite } K_H. \tag{S-87}$$

S-1.13.3 Effective affinity constant and adsorption free energy

An effective affinity constant is defined from the half-loading concentration $C_{1/2}$ (the concentration at which $q_e = Q_{\max}/2$). Setting $q_e = Q_{\max}/2$ gives $\varepsilon_{1/2} = E_a(\ln 2)^{1/n_{DA}}$, and

$$C_{1/2} = \frac{1}{\exp(\varepsilon_{1/2}/RT) - 1}. \tag{S-88}$$

The effective affinity constant and energy are

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$K_{\text{aff}} = 1/C_{1/2}, \qquad K^*_{\text{aff}} = C^0/C_{1/2}, \qquad E_{\text{aff}} = RT\ln(C^0/C_{1/2}). \tag{S-89}$$

S-1.14 Hill isotherm

The Hill isotherm is

$$q_e = Q_{\max} \frac{C_e^{n_H}}{K_D + C_e^{n_H}}, \tag{S-90}$$

where n_H is the Hill coefficient and K_D is the dissociation constant. When $n_H = 1$, the model reduces to Langmuir with $K_L = 1/K_D$.

S-1.14.1 High concentration limit: finite $Q_{\max}$

As $C_e \to \infty$, $C_e^{n_H} \gg K_D$, so

$$\lim_{C_e \to \infty} q_e = Q_{\max}. \tag{S-91}$$

Capacity-limit result

Hill has a finite saturation capacity:

$$\lim_{C_e \to \infty} q_e = Q_{\max}. \tag{S-92}$$

Report $Q_{\max}$ as capacity and interpret n_H separately as the cooperativity or heterogeneity-shape parameter.

S-1.14.2 Low concentration limit: conditional Henry constant

For $C_e^{n_H} \ll K_D$,

$$q_e \approx \frac{Q_{\max}}{K_D} C_e^{n_H}. \tag{S-93}$$

A finite K_H exists only when $n_H = 1$:

$$K_H = Q_{\max}/K_D \quad (n_H = 1). \tag{S-94}$$

For $n_H \neq 1$, the initial slope is concentration dependent:

$$K_H(C_e) = \frac{Q_{\max}}{K_D}\, C_e^{n_H - 1}. \tag{S-95}$$

S-1.14.3 Affinity constant and adsorption free energy

The effective affinity constant with units $[C]^{-1}$ is $K_{\text{aff}} = K_D^{-1/n_H}$. Therefore,

$$K_{\text{aff}}^{*} = K_D^{-1/n_H}\, C^0, \tag{S-96}$$

and the adsorption free energy is

Standard-state conversion

Use this form to convert model-specific constants into a dimensionless affinity and comparable adsorption energy. Always report the chosen standard concentration C^0.

$$E_{\text{aff}} = RT\left(-\frac{1}{n_H}\ln K_D + \ln C^0\right). \tag{S-97}$$

In the limit $n_H = 1$, this reduces to the Langmuir expression $E_{\text{aff}} = RT\ln(K_L\, C^0)$ with $K_L = 1/K_D$.

S-2 Error metrics and information criteria

What this section is for

Error metrics answer a statistical question: "which curve follows the data best?" They do not by themselves prove that a model is physically meaningful. A good workflow uses both: residual statistics first, physical descriptor checks second.

S-2.1 Residual error metrics

Let the residual at point i be

$$r_i = q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}}(\boldsymbol{\theta}; C_{e,i}), \qquad \mathbf{r} = (r_1, \ldots, r_n)^\top, \tag{S-98}$$

where q_i^{exp} and q_i^{pred} are the experimental and predicted adsorption capacities, n is the number of data points, p is the number of fitted parameters, and

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} r_i^2 = \arg\min_{\boldsymbol{\theta}} \|\mathbf{r}\|_2^2. \tag{S-99}$$

Equation (S-99) is the unweighted non-linear least-squares problem assumed throughout the main manuscript and implemented in the companion notebooks. If the experimental variance is known to be strongly non-uniform, a weighted objective $\sum_i w_i r_i^2$ with $w_i \propto 1/\sigma_i^2$ may be preferable, but all metrics below are given for the unweighted case so that different models remain directly comparable.

For convenience, define the sample mean capacity

$$\bar{q} = \frac{1}{n} \sum_{i=1}^{n} q_i^{\mathrm{exp}}. \tag{S-100}$$

S-2.1.1 Weighted residuals and derived error scales

If the experimental uncertainty at each point is known or can be estimated, the weighted least-squares objective is

$$\mathrm{WSSE} = \sum_{i=1}^{n} w_i r_i^2, \qquad w_i = \frac{1}{\sigma_i^2}, \tag{S-101}$$

where σ_i is the standard deviation of the measurement at $C_{e,i}$. In that case the natural likelihood-based discrepancy is the chi-squared statistic

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{r_i}{\sigma_i} \right)^2, \tag{S-102}$$

with reduced chi-squared

$$\chi^2_{\mathrm{red}} = \frac{\chi^2}{n-p}. \tag{S-103}$$

When all σ_i are equal, χ^2 is proportional to SSE and therefore does not alter model ranking.

Two additional scalar error scales are often useful:

$$\mathrm{MSE} = \frac{\mathrm{SSE}}{n}, \qquad \mathrm{RMSE} = \sqrt{\frac{\mathrm{SSE}}{n}}, \tag{S-104}$$

and the residual variance estimate used in parameter-error propagation is

$$s_r^2 = \frac{\mathrm{SSE}}{n-p}. \tag{S-105}$$

Equation (S-105) is the unbiased variance estimator for the residuals after accounting for the p fitted parameters.

S-2.1.2 Absolute error metrics

The sum of squared errors (SSE) is the objective function minimised during non-linear regression:

$$\mathrm{SSE} = \sum_{i=1}^{n} (q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}})^2. \tag{S-106}$$

The coefficient of determination R^2 gives the fraction of variance explained by the model:

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}, \qquad \mathrm{SST} = \sum_i (q_i^{\mathrm{exp}} - \bar{q})^2. \tag{S-107}$$

The adjusted R^2 penalises for the number of fitted parameters, preventing overfitting from being rewarded:

$$R^2_{\mathrm{adj}} = 1 - \frac{n-1}{n-p-1}(1 - R^2). \tag{S-108}$$

The sum of absolute errors (EABS) provides a measure less sensitive to outliers than SSE:

$$\mathrm{EABS} = \sum_i |q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}}|. \tag{S-109}$$

The mean absolute residual (RESID) normalises EABS by the number of data points, giving a typical per-point deviation in the same units as q_e:

$$\mathrm{RESID} = \frac{1}{n} \sum_i |q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}}|. \tag{S-110}$$

SSE is sensitive to large deviations because the residuals are squared, whereas EABS and RESID remain in the original q_e units and are less dominated by a small number of outliers. For non-linear isotherms, R^2 and R^2_{adj} are descriptive summaries of fit quality; they should not be used as the sole basis for model selection because they do not encode any physical penalty for additional parameters.

S-2.1.3 Relative error metrics

The average relative error in percent (ARED) normalises each residual by the experimental value, giving equal weight to low- and high-concentration data. This metric is particularly informative when data span several orders of magnitude:

$$\mathrm{ARED} = \frac{100}{n} \sum_i \left| \frac{q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}}}{q_i^{\mathrm{exp}}} \right|. \tag{S-111}$$

The median percentage squared error (MPSED) replaces the mean with a median, making it robust to individual outliers:

$$\mathrm{MPSED} = 100 \times \mathrm{median} \left[\left(\frac{q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}}}{q_i^{\mathrm{exp}}} \right)^2 \right]. \tag{S-112}$$

The HYBRID error function combines squared residuals with a $1/q_i^{\mathrm{exp}}$ weight, amplifying deviations at low concentrations where q_i^{exp} is small. It also corrects for degrees of freedom $(n-p)$:

$$\mathrm{HYBRID} = \frac{100}{n-p} \sum_i \frac{(q_i^{\mathrm{exp}} - q_i^{\mathrm{pred}})^2}{q_i^{\mathrm{exp}}}. \tag{S-113}$$

ARED, MPSED, and HYBRID require $q_i^{\mathrm{exp}} > 0$. This is consistent with adsorption datasets after removal of zero-capacity points, which otherwise would introduce singular denominators in the relative-error metrics.

Recommended combination. No single metric captures all aspects of fit quality. We recommend reporting at least SSE (or R^2_{adj}) for absolute quality, ARED for relative quality, HYBRID for low-concentration sensitivity, and AIC_c (Section S-2.2) for model-complexity penalisation.

S-2.1.4 Covariance-matrix errors, confidence intervals, and bands

Let the Jacobian of the fitted model be

$$J_{ij} = \left. \frac{\partial q^{\mathrm{pred}}(C_{e,i}; \boldsymbol{\theta})}{\partial \theta_j} \right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}, \tag{S-114}$$

and denote by $\mathbf{J}$ the $n \times p$ matrix with entries J_{ij}. For unweighted non-linear least squares, the parameter covariance matrix is approximated by

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \approx s_r^2 (\mathbf{J}^\mathsf{T}\mathbf{J})^{-1}, \tag{S-115}$$

where s_r^2 is given by Eq. (S-105). In the weighted case the corresponding approximation becomes

$$\mathrm{Cov}_w(\hat{\boldsymbol{\theta}}) \approx (\mathbf{J}^\mathsf{T}\mathbf{W}\mathbf{J})^{-1}, \qquad \mathbf{W} = \mathrm{diag}(w_1, \dots, w_n). \tag{S-116}$$

The standard error of parameter θ_j is therefore

$$\mathrm{SE}(\hat{\theta}_j) = \sqrt{\left[\mathrm{Cov}(\hat{\boldsymbol{\theta}})\right]_{jj}}, \tag{S-117}$$

and a two-sided $(1-\alpha)$ confidence interval is

$$\hat{\theta}_j \pm t_{1-\alpha/2,\, n-p}\, \mathrm{SE}(\hat{\theta}_j), \tag{S-118}$$

where $t_{1-\alpha/2,\, n-p}$ is the Student-t quantile with $n-p$ degrees of freedom.

For any predicted uptake at concentration C_e, define the sensitivity vector

$$\mathbf{g}(C_e) = \nabla_{\boldsymbol{\theta}} q^{\mathrm{pred}}(C_e; \boldsymbol{\theta})\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}. \tag{S-119}$$

The covariance-based prediction variance is then

$$\mathrm{Var}[\hat{q}(C_e)] \approx \mathbf{g}(C_e)^{\mathsf{T}} \mathrm{Cov}(\hat{\boldsymbol{\theta}}) \mathbf{g}(C_e), \tag{S-120}$$

which gives the approximate pointwise confidence band

$$\hat{q}(C_e) \pm z_{1-\alpha/2} \sqrt{\mathrm{Var}[\hat{q}(C_e)]}, \tag{S-121}$$

with $z_{1-\alpha/2}$ the standard normal quantile. This is the covariance-matrix confidence-band construction used in the companion notebooks for visual comparison of fitted curves.

S-2.1.5 Residual diagnostics, bootstrap, and noise sensitivity

Residual metrics alone do not show whether the fitted model is structurally adequate. The residual sequence $\{r_i\}_{i=1}^{n}$ should therefore be inspected against C_e (or against fitted values) to verify that no systematic curvature remains. For a well-specified model, the residuals should fluctuate around zero without concentration-dependent trends.

Parameter and descriptor uncertainty can be propagated by residual bootstrap. Let

$$r_i^{\mathrm{c}} = r_i - \bar{r}, \qquad \bar{r} = \frac{1}{n}\sum_{i=1}^{n} r_i, \tag{S-122}$$

and generate bootstrap datasets

$$q_{i,b}^{*} = q_i^{\mathrm{pred}}(\hat{\boldsymbol{\theta}}; C_{e,i}) + r_{j_b(i)}^{\mathrm{c}}, \qquad j_b(i) \in \{1, \dots, n\}, \tag{S-123}$$

where the indices $j_b(i)$ are sampled with replacement. Re-fitting the model to each bootstrap dataset gives a family of parameter vectors $\hat{\boldsymbol{\theta}}^{(b)}$ and therefore a family of descriptors $d^{(b)} \in \{Q_{\mathrm{max}}, E_{\mathrm{ads}}, \sigma_H\}$. The bootstrap standard deviation of any scalar descriptor d is

$$s_d^{\mathrm{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(d^{(b)} - \bar{d}\right)^2}, \qquad \bar{d} = \frac{1}{B}\sum_{b=1}^{B} d^{(b)}. \tag{S-124}$$

Percentile intervals from the empirical distribution of $d^{(b)}$ are a practical way to report confidence intervals on the extracted descriptors. Explicitly, the bootstrap percentile interval for any scalar quantity x is

$$\mathrm{CI}_{1-\alpha}^{\mathrm{boot}}(x) = \left[Q_{\alpha/2}\big(x^{(b)}\big)\,,\ Q_{1-\alpha/2}\big(x^{(b)}\big)\right], \tag{S-125}$$

where $Q_\beta(x^{(b)})$ is the empirical β-quantile of the bootstrap sample.

Sensitivity to measurement noise can be examined by perturbing the fitted response with controlled relative noise levels,

$$q_{i,b}^{(\eta)} = q_i^{\mathrm{pred}}(\hat{\boldsymbol{\theta}}; C_{e,i})\left(1 + \eta z_{i,b}\right), \qquad z_{i,b} \sim \mathcal{N}(0,1), \tag{S-126}$$

with $\eta = 0.01$, 0.05, or 0.10 for 1%, 5%, and 10% relative noise, respectively. Parameters or descriptors whose mean shift under 5% noise exceeds their bootstrap standard deviation should be regarded as weakly constrained by the available data. For a given descriptor d, the relative noise-sensitivity index can be written as

$$\delta_d^{(\eta)} = \frac{\left|\bar{d}^{(\eta)} - \hat{d}\right|}{\max\left(|\hat{d}|, \varepsilon\right)}, \qquad \bar{d}^{(\eta)} = \frac{1}{B_\eta}\sum_{b=1}^{B_\eta} d_b^{(\eta)}, \tag{S-127}$$

where $\hat{d}$ is the descriptor from the original fit and ε is a small numerical safeguard for near-zero quantities. This makes explicit the error-treatment criterion used in the notebook workflow when testing parameter robustness to synthetic perturbations.

S-2.2 Information criteria for model selection

Assuming independent Gaussian residuals with constant variance σ^2, the log-likelihood of a fitted model is

$$\log L(\hat{\boldsymbol{\theta}}, \hat{\sigma}^2) = -\frac{n}{2}\left[\ln(2\pi) + 1 + \ln\left(\frac{\mathrm{SSE}}{n}\right)\right], \tag{S-128}$$

where the maximum-likelihood variance estimate is $\hat{\sigma}^2 = \mathrm{SSE}/n$. Up to an additive constant common to all models fitted to the same dataset, minimising the negative log-likelihood is equivalent to minimising $n\ln(\mathrm{SSE}/n)$. This is the basis of the AIC, AIC_c, and BIC expressions below.

$$\mathrm{AIC} = n\ln\left(\frac{\mathrm{SSE}}{n}\right) + 2p, \tag{S-129}$$

$$\mathrm{AIC}_c = \mathrm{AIC} + \frac{2p(p+1)}{n-p-1}, \tag{S-130}$$

$$\mathrm{BIC} = n\ln\left(\frac{\mathrm{SSE}}{n}\right) + p\ln n. \tag{S-131}$$

The model with the *lowest* AIC_c (or BIC) is preferred. Interpretation of differences: $\Delta\mathrm{AIC} < 2$: substantial support for both models; $4 < \Delta\mathrm{AIC} < 7$: considerably less support for the higher-AIC model; $\Delta\mathrm{AIC} > 10$: essentially no support. Only models fitted to the same response variable, using the same data points and the same error definition, should be compared through information criteria.

Akaike weights normalise the relative likelihood of each model:

$$w_i = \frac{\exp(-\Delta\mathrm{AIC}_i/2)}{\sum_j \exp(-\Delta\mathrm{AIC}_j/2)}. \tag{S-132}$$

These weights satisfy $\sum_i w_i = 1$ and admit a natural model-averaged prediction,

$$\hat{q}_{\mathrm{avg}}(C_e) = \sum_{i=1}^{M} w_i\, \hat{q}_i(C_e), \tag{S-133}$$

or, analogously, a model-averaged descriptor $d_{\mathrm{avg}} = \sum_i w_i d_i$ when several models remain within $\Delta\mathrm{AIC}_c < 2$.

S-3 Model selection

Practical selection rule

Choose the simplest model that gives acceptable residuals and physically meaningful descriptors. A lower AIC_c is useful, but it should not overrule impossible quantities such as a fake $Q_{\max}$ from a non-saturating equation.

S-3.1 Ranking procedure

1. Compute AIC_c for all candidate models; rank by $\Delta\mathrm{AIC}$.
2. For nested pairs (Langmuir $\subset$ Sips, Langmuir $\subset$ Double Langmuir, Langmuir $\subset$ Redlich–Peterson), apply the F-test:

$$F = \frac{(\mathrm{SSE}_1 - \mathrm{SSE}_2)/(p_2 - p_1)}{\mathrm{SSE}_2/(n - p_2)}. \tag{S-134}$$

If $F > F_\alpha(p_2 - p_1,\, n - p_2)$, the more complex model provides a statistically significant improvement at confidence level α.

3. Inspect residual plots for systematic patterns.
4. Compute Akaike weights (Eq. S-132).
5. If multiple models have $\Delta\text{AIC} < 2$, report all and consider model averaging.

For completeness, the associated upper-tail p-value is

$$p_F = 1 - F_{\nu_1,\nu_2}(F_{\text{obs}}), \qquad \nu_1 = p_2 - p_1, \qquad \nu_2 = n - p_2, \tag{S-135}$$

where F_{ν_1,ν_2} is the cumulative distribution function of the F distribution. The F-test is meaningful only when the two candidate models are truly nested, fitted to the same dataset, and optimised with the same residual definition.

S-3.2 Energy-distribution overlap

The overlap coefficient between two normalised energy distributions $\hat{f}_A(E)$ and $\hat{f}_B(E)$ quantifies their similarity:

$$\text{OVL} = \int_{-\infty}^{\infty} \min\big[\hat{f}_A(E),\ \hat{f}_B(E)\big]\,\mathrm{d}E. \tag{S-136}$$

For approximately Gaussian distributions:

$$\text{OVL} \approx 2\,\Phi\left(-\frac{|E_c^A - E_c^B|}{2\bar{\sigma}}\right), \qquad \bar{\sigma} = \sqrt{\frac{\sigma_A^2 + \sigma_B^2}{2}}. \tag{S-137}$$

For arbitrary distributions represented on a numerical energy grid $\{E_g\}_{g=1}^{G}$, the overlap is evaluated by quadrature as

$$\text{OVL}_{\text{num}} \approx \sum_{g=1}^{G-1} \min\big[\hat{f}_A(E_g), \hat{f}_B(E_g)\big]\,\Delta E_g, \qquad \Delta E_g = E_{g+1} - E_g, \tag{S-138}$$

after enforcing unit-area normalisation on each distribution. This numerical form is the appropriate one for asymmetric or non-standard families such as Tóth, Jovanovic, Redlich–Peterson, or Radke–Prausnitz, for which a closed-form overlap is generally not available.

Interpretation.

- OVL > 0.8: models are practically indistinguishable — prefer the simpler one.
- $0.3 <$ OVL < 0.8: partial overlap; use AIC/BIC to discriminate.
- OVL < 0.3: distributions describe different physics.

S-3.3 Specific comparison criteria

S-3.3.1 Langmuir vs. Sips.

Model-selection rule

Use this rule only after checking the residual plots and information criteria. It is a practical decision aid that connects statistical fit quality with the physical descriptor space used in the manuscript.

$$\text{Accept Langmuir if}\quad \sigma_H^{(\mathrm{Sips})} = \frac{\pi RT}{\sqrt{3}}\sqrt{\frac{1}{m^2} - 1} \leq \alpha\, RT, \tag{S-139}$$

with $\alpha \approx 1$ as a practical screening value. Smaller values of α are stricter and collapse toward the Langmuir limit $m \to 1$, where $\sigma_H \to 0$.

S-3.3.2 Langmuir vs. Double Langmuir.

Model-selection rule

Use this rule only after checking the residual plots and information criteria. It is a practical decision aid that connects statistical fit quality with the physical descriptor space used in the manuscript.

$$\text{Accept Double Langmuir if} \quad \Delta E_{12} = RT \left| \ln \frac{K_1}{K_2} \right| > \beta\, RT, \tag{S-140}$$

with $\beta \approx 2$–3 ($K_1/K_2 > 7.4$–20).

S-3.3.3 Sips vs. Double Langmuir.

Model-selection rule

Use this rule only after checking the residual plots and information criteria. It is a practical decision aid that connects statistical fit quality with the physical descriptor space used in the manuscript.

$$\text{Prefer Sips over DL if} \quad \Delta E_{12} < 2\, \sigma_H^{(\text{Sips})}. \tag{S-141}$$

S-3.4 Decision tree

The model-selection decision tree proceeds in three phases: (i) physical screening of the data shape, (ii) AIC ranking among applicable models, and (iii) nested-model tests with physical validation. Figure S-1 gives the complete flowchart.

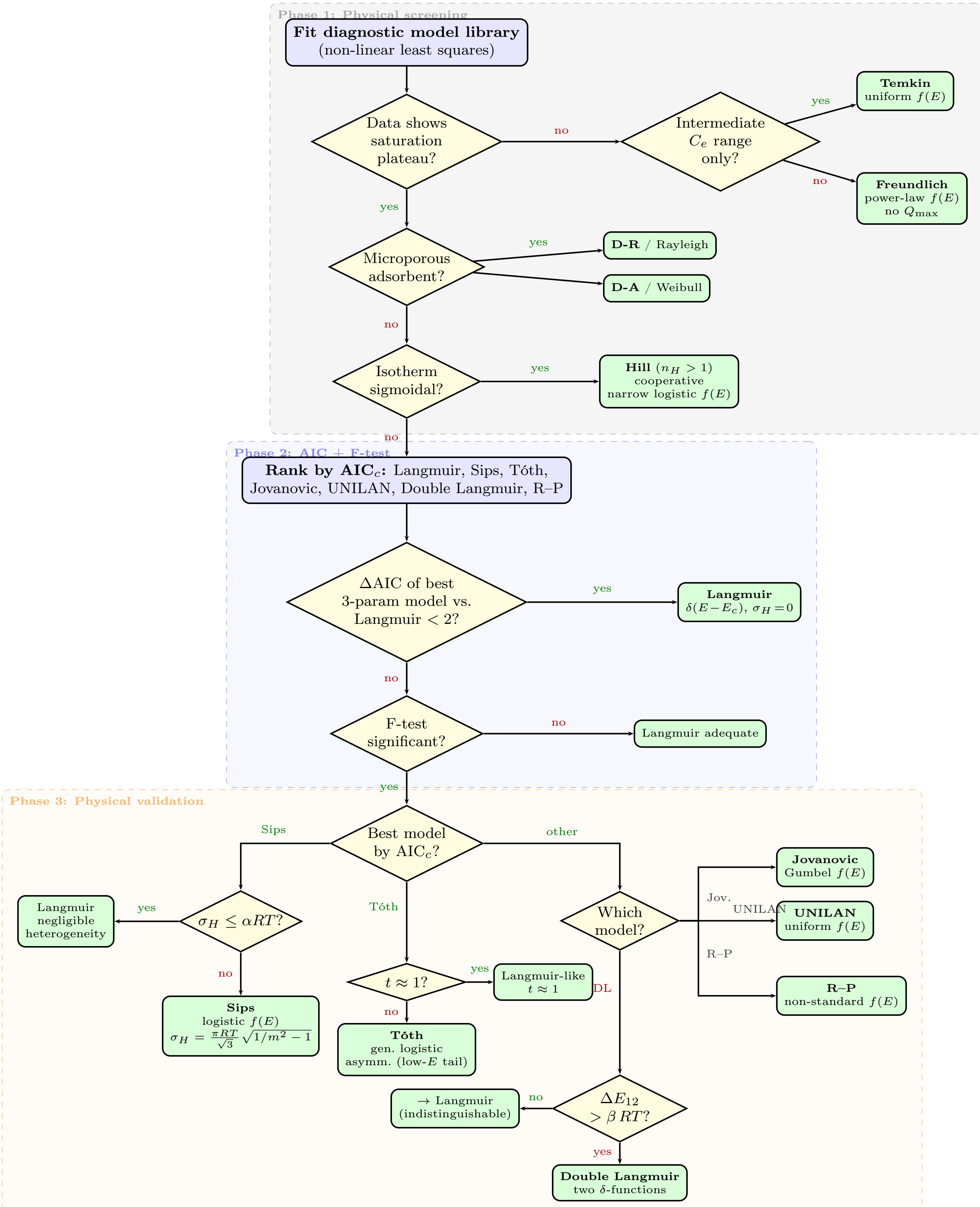

Figure S-1: Complete decision tree for isotherm model selection. **Phase 1** (top): physical screening based on data shape — saturation plateau, micropore filling, sigmoidal curvature. **Phase 2** (middle): AIC$_c$ ranking and F-test for all saturating models. **Phase 3** (bottom): pairwise tests combined with physical criteria (σ_H, ΔE_{12}, t, β). Green terminal boxes show the accepted model and its energy-distribution family.

S-4 Energy distributions of adsorption isotherms

S-4.1 General framework

A heterogeneous surface is described as a collection of independent Langmuir sites, each with adsorption energy E. The macroscopic isotherm is the integral:

$$q_e = Q_{\max} \int_0^\infty \frac{K(E)\, C_e}{1 + K(E)\, C_e}\, f(E)\, \mathrm{d}E, \tag{S-142}$$

where $K(E) = K_0 \exp(E/RT)$ and $f(E)$ is the normalised site-energy distribution. Different choices of $f(E)$ yield different macroscopic isotherms.

In the *condensation approximation*, the substitution

$$C_e = C^* \exp(-E/RT) \tag{S-143}$$

maps the isotherm $q(C_e)$ into $q(E)$, and the energy distribution is recovered by differentiation:

$$f(E) = -\frac{\mathrm{d}q}{\mathrm{d}E}. \tag{S-144}$$

Width convention

Use σ_E for the apparent width obtained from the mathematical energy distribution. Use σ_H for the intrinsic heterogeneity descriptor reported in the final tables. For logistic-family models (Sips, Koble–Corrigan, Hill), the Langmuir thermal width is

$$\sigma_T = \frac{\pi}{\sqrt{3}} RT, \qquad \sigma_H = \sqrt{\max\{\sigma_E^2 - \sigma_T^2,\, 0\}}.$$

Thus Langmuir and the homogeneous limit of Sips give the same reported value: $\sigma_H = 0$.

S-4.2 Summary of all energy distributions

Table S-1 collects the distribution family, symmetry, and width parameter σ_E for every isotherm model.

Table S-1: Energy-distribution families implied by each isotherm model. All distributions are expressed in the reduced energy $u = (E - E_c)/(RT)$. The standard deviation σ_E is given in units of RT (≈ 2.478 kJ/mol at $T = 298$ K).

Model	**Distribution** $f(E)$	**Symmetry**	$\boldsymbol{\sigma_E/(RT)}$	**Fit parameters**
Langmuir	Dirac $\delta(E - E_c)$	—	0	K_L, $Q_{\max}$
Double Langmuir	Two Dirac δ	—	$\dfrac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2} \dfrac{\lvert\Delta E_{12}\rvert}{RT}$	K_1, K_2, Q_1, Q_2
Sips (L–F)	Logistic	Symmetric	$\pi/(m\sqrt{3})$	K_S, m, $Q_{\max}$
Tóth	Gen. logistic (IV)	Asymmetric	$> \pi/\sqrt{3}$ ($t < 1$)	b, t, $q_{\max}$
Jovanovic	Gumbel (EV-I)	Asymmetric	$\pi/\sqrt{6} \approx 1.28$	b, $q_{\max}$
Temkin	Uniform (rect.)	Symmetric	$\dfrac{b_T Q_{\max}}{2\sqrt{3}\, RT}$	a_T, b_T
UNILAN	Uniform (rect.)	Symmetric	$\dfrac{\ln(b_{\max}/b_{\min})}{2\sqrt{3}}$	$b_{\max}, b_{\min}, q_{\max}$
Freundlich	Power law	—	∞	K_F, $1/n$
Redlich–Peterson	Non-standard	Asymm. ($\beta < 1$)	—	K_{RP}, a_{RP}, β
Koble–Corrigan	$\equiv$ Sips	Symmetric	$\pi n/\sqrt{3}$	$A_{KC}, B_{KC}, 1/n$
Radke–Prausnitz	Non-standard	Asymmetric	—	a_{RP}, r_{RP}, p
D-R	Rayleigh	Asymmetric	$0.463\, E_a/RT$	$Q_{\max}$, E_a
D-A	Weibull	Asymmetric	$\frac{E_a}{RT}\sqrt{\Gamma(1+\frac{2}{n}) - \Gamma(1+\frac{1}{n})^2}$	$Q_{\max}, E_a, n_{DA}$
Hill	$\equiv$ Sips	Symmetric	$\pi/(\sqrt{3}\, n_H)$	$Q_{\max}, K_D, n_H$

S-4.3 Numerical values of σ_E at $T = 298$ K

Table S-2 gives numerical values of σ_E for representative parameter choices at $T = 298$ K ($RT = 2.478$ kJ mol^{-1}).

Table S-2: Numerical values of σ_E at $T = 298$ K ($RT = 2.478$ kJ mol^{-1}) for representative parameter values.

Model	**Parameter**	$\boldsymbol{\sigma_E/(RT)}$	$\boldsymbol{\sigma_E}$ (kJ/mol)
Sips	$m = 1.0$	1.81	4.49
Sips	$m = 0.7$	2.59	6.41
Sips	$m = 0.5$	3.63	8.99
Sips	$m = 0.33$	5.49	13.60
Tóth	$t = 0.7$	2.17	5.38
Tóth	$t = 0.5$	2.65	6.56
Tóth	$t = 0.3$	3.51	8.70
Jovanovic	—	1.28	3.18
Temkin/UNILAN	$s = 2\,RT$	1.15	2.86
Temkin/UNILAN	$s = 3\,RT$	1.73	4.29
Temkin/UNILAN	$s = 4\,RT$	2.31	5.72
Hill	$n_H = 0.5$	3.63	8.99
Hill	$n_H = 1.0$	1.81	4.49
Hill	$n_H = 2.0$	0.91	2.24
Hill	$n_H = 4.0$	0.45	1.12
D-R	$E_a = 10$ kJ/mol	1.87	4.63
D-R	$E_a = 20$ kJ/mol	3.74	9.27

How to use this table

The values in Table S-2 are useful for understanding the shape of the mathematical energy distributions. For final material comparison, use the reporting convention in Table S-4; in particular, logistic-family models are reported with σ_H.

S-4.4 Individual energy distributions and σ_E derivations

The condensation approximation (Section S-4.1) is applied to each isotherm by substituting $C_e = C^* \mathrm{e}^{-E/RT}$ and computing $f(E) = -(1/Q_{\max})\,\mathrm{d}q/\mathrm{d}E$. The standard deviation σ_E is then obtained from the variance of the resulting distribution family.

S-4.4.1 Langmuir

Exact result. The Langmuir isotherm is obtained *exactly* by setting $f(E) = \delta(E - E_c)$ in Eq. (S-142):

$$Q_{\max} \int_0^{\infty} \frac{K(E)\, C_e}{1 + K(E)\, C_e}\, \delta(E - E_c)\, \mathrm{d}E = Q_{\max}\, \frac{K_L\, C_e}{1 + K_L\, C_e}, \qquad K_L = K_0 \mathrm{e}^{E_c/RT}. \tag{S-145}$$

No condensation approximation is needed: the Langmuir isotherm is *exact* for a homogeneous surface ($\sigma_E = 0$).

Characteristic energy and width.

$$E_c = RT \ln(K_L\, C^*), \qquad \sigma_E = 0. \tag{S-146}$$

S-4.4.2 Double Langmuir

Distribution. The double-Langmuir isotherm follows from a bimodal discrete distribution:

$$f(E) = \frac{Q_1}{Q_{\max}}\, \delta(E - E_1) + \frac{Q_2}{Q_{\max}}\, \delta(E - E_2), \qquad Q_{\max} = Q_1 + Q_2, \tag{S-147}$$

with $E_i = RT \ln(K_i\, C^*)$.

Derivation of σ_E. For a two-point distribution with weights $w_i = Q_i / Q_{\max}$:

$$\langle E \rangle = w_1 E_1 + w_2 E_2, \tag{S-148}$$

$$\sigma_E^2 = w_1 (E_1 - \langle E \rangle)^2 + w_2 (E_2 - \langle E \rangle)^2 = w_1 w_2\, (E_1 - E_2)^2, \tag{S-149}$$

therefore

$$\sigma_E = \frac{\sqrt{Q_1 Q_2}}{Q_1 + Q_2}\, |\Delta E_{12}|, \qquad \Delta E_{12} = E_1 - E_2. \tag{S-150}$$

For equal sites ($Q_1 = Q_2$): $\sigma_E = |\Delta E_{12}|/2$.

S-4.4.3 Sips (Langmuir–Freundlich)

Starting isotherm.

$$q_e = Q_{\max} \frac{K_S C_e^m}{1 + K_S C_e^m}, \qquad 0 < m \leq 1. \tag{S-151}$$

Condensation substitution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$ and defining $E_c = (RT/m) \ln[K_S (C^*)^m] = RT \ln(K_S^{1/m} C^*)$:

$$q(E) = \frac{Q_{\max}}{1 + \mathrm{e}^{m(E-E_c)/RT}}. \tag{S-152}$$

Differentiation.

$$\frac{\mathrm{d}q}{\mathrm{d}E} = -Q_{\max} \frac{(m/RT)\, \mathrm{e}^{m(E-E_c)/RT}}{\left[1 + \mathrm{e}^{m(E-E_c)/RT}\right]^2}. \tag{S-153}$$

Distribution.

$$f(E) = \frac{Q_{\max}\, (m/RT)\, \mathrm{e}^{m(E-E_c)/RT}}{\left[1 + \mathrm{e}^{m(E-E_c)/RT}\right]^2}. \tag{S-154}$$

This is the **logistic** probability density (times $Q_{\max}$) with location E_c and scale $s = RT/m$.

Width. The variance of the logistic distribution with scale s is $\pi^2 s^2/3$, so

$$\sigma_E = \frac{\pi}{\sqrt{3}} \frac{RT}{m}. \tag{S-155}$$

For $m = 1$ (Langmuir): $\sigma_E = (\pi/\sqrt{3})\, RT \approx 1.81\, RT$. As $m \to 0$: $\sigma_E \to \infty$ (Freundlich limit).

S-4.4.4 Tóth

Starting isotherm.

$$q_e = \frac{q_{\max}\, b\, C_e}{\left[1 + (bC_e)^t\right]^{1/t}}, \qquad 0 < t \leq 1. \tag{S-156}$$

Condensation substitution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$, $E_c = RT\ln(bC^*)$, $u = (E - E_c)/RT$:

$$q(E) = q_{\max} \frac{\mathrm{e}^{-u}}{\left[1 + \mathrm{e}^{-tu}\right]^{1/t}}. \tag{S-157}$$

Differentiation. Applying the product rule and collecting terms:

$$\frac{\mathrm{d}q}{\mathrm{d}E} = -\frac{q_{\max}}{RT} \frac{\mathrm{e}^{-u}}{\left[1 + \mathrm{e}^{-tu}\right]^{(1+t)/t}}, \qquad u = \frac{E - E_c}{RT}. \tag{S-158}$$

Distribution.

$$f(E) = \frac{q_{\max}}{RT} \frac{\mathrm{e}^{-u}}{[1 + \mathrm{e}^{-tu}]^{(1+t)/t}}, \qquad u = \frac{E - E_c}{RT}. \tag{S-159}$$

This is the **generalised logistic type IV**, asymmetric with a heavier tail toward *low* adsorption energies for $t < 1$. For $t = 1$ it reduces to the standard logistic (Langmuir limit).

Width. No closed-form σ_E exists for general t; numerical integration is required. $\sigma_E > (\pi/\sqrt{3})\, RT$ for all $t < 1$, diverging as $t \to 0$.

S-4.4.5 Jovanovic

Starting isotherm.

$$q_e = q_{\max}\left[1 - \exp(-bC_e)\right]. \tag{S-160}$$

Condensation substitution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$, $E_c = RT\ln(bC^*)$, $u = (E - E_c)/RT$:

$$q(E) = q_{\max}\left[1 - \exp\left(-\mathrm{e}^{-u}\right)\right]. \tag{S-161}$$

Differentiation.

$$\frac{\mathrm{d}q}{\mathrm{d}E} = -\frac{q_{\max}}{RT}\, \mathrm{e}^{-u} \exp\left(-\mathrm{e}^{-u}\right). \tag{S-162}$$

Distribution.

$$f(E) = \frac{q_{\max}}{RT}\, \mathrm{e}^{-u} \exp\left(-\mathrm{e}^{-u}\right), \qquad u = \frac{E - E_c}{RT}. \tag{S-163}$$

This is the **Gumbel (extreme-value type I)** distribution with location E_c and scale RT. The mean is $\langle E \rangle = E_c + \gamma\, RT$ ($\gamma \approx 0.5772$, Euler–Mascheroni constant). The distribution is asymmetric with a longer tail toward *high* energies.

Width. The variance of the Gumbel distribution with scale β is $\pi^2\beta^2/6$:

$$E_c = RT\ln(bC^*), \qquad \sigma_E = \frac{\pi}{\sqrt{6}}\, RT \approx 1.283\, RT. \tag{S-164}$$

The width is independent of the affinity parameter b and is fixed solely by the model functional form.

S-4.4.6 Temkin

Starting isotherm.

$$q_e = \frac{RT}{b_T}\ln(a_T C_e). \tag{S-165}$$

(Valid over the intermediate concentration range where the log-linear approximation holds; implicitly defines $[E_{\min}, E_{\max}]$.)

Condensation substitution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$:

$$q(E) = \frac{1}{b_T}\big[RT\ln(a_T C^*) - E\big]. \tag{S-166}$$

Distribution.

$$f(E) = -\frac{1}{Q_{\max}}\frac{\mathrm{d}q}{\mathrm{d}E} = \frac{Q_{\max}}{b_T Q_{\max}} = \frac{1}{b_T} = \text{const}, \qquad E_{\min} \le E \le E_{\max}. \tag{S-167}$$

This is a **uniform (rectangular)** distribution over $[E_{\min}, E_{\max}]$, where $E_{\max} = RT\ln(a_T C^*)$ and $E_{\max} - E_{\min} = b_T Q_{\max}$.

Width. For a uniform distribution on $[a, b]$, $\sigma = (b - a)/(2\sqrt{3})$:

$$\sigma_E = \frac{E_{\max} - E_{\min}}{2\sqrt{3}} = \frac{b_T\, Q_{\max}}{2\sqrt{3}}. \tag{S-168}$$

S-4.4.7 UNILAN

Exact integration. UNILAN retains the full Langmuir kernel and integrates over a uniform distribution $f(E) = Q_{\max}/(2s)$ on $[E_c - s,\ E_c + s]$ analytically:

$$\begin{aligned} q_e &= \frac{Q_{\max}}{2s} \int_{E_c - s}^{E_c + s} \frac{K_0 \mathrm{e}^{E/RT}\, C_e}{1 + K_0 \mathrm{e}^{E/RT}\, C_e}\, \mathrm{d}E \\ &= \frac{Q_{\max}\, RT}{2s} \ln \frac{1 + b_{\max}\, C_e}{1 + b_{\min}\, C_e}, \qquad b_{\max\,/\,\min} = K_0\, \mathrm{e}^{(E_c \pm s)/RT}. \end{aligned} \tag{S-169}$$

Distribution and width. By construction:

$$f(E) = \frac{Q_{\max}}{2s} = \text{const}, \qquad s = \frac{RT}{2} \ln \frac{b_{\max}}{b_{\min}}. \tag{S-170}$$

The standard deviation of the uniform distribution on $[E_c - s,\ E_c + s]$ is $s/\sqrt{3}$:

$$\sigma_E = \frac{s}{\sqrt{3}} = \frac{RT}{2\sqrt{3}} \ln \frac{b_{\max}}{b_{\min}}. \tag{S-171}$$

S-4.4.8 Freundlich

Starting isotherm.

$$q_e = K_F C_e^{1/n}, \qquad 0 < 1/n < 1. \tag{S-172}$$

Condensation substitution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$:

$$q(E) = K_F (C^*)^{1/n}\, \mathrm{e}^{-E/(nRT)}. \tag{S-173}$$

Distribution.

$$f(E) = \frac{K_F (C^*)^{1/n}}{nRT} \, \mathrm{e}^{-E/(nRT)}. \tag{S-174}$$

This one-sided exponential does *not* normalise to a finite area on $[0, \infty)$, reflecting the absence of a finite $Q_{\max}$.

Width. $\sigma_E = \infty$. The Freundlich model cannot be interpreted as arising from a physically normalised site-energy distribution; it is a purely empirical fitting equation valid only over a limited concentration window.

S-4.4.9 Redlich–Peterson

Starting isotherm.

$$q_e = \frac{K_{RP}\, C_e}{1 + a_{RP}\, C_e^{\beta}}, \qquad 0 < \beta \leq 1. \tag{S-175}$$

Condensation substitution and distribution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$ and differentiating:

$$f(E) = \frac{K_{RP}\, C^* \mathrm{e}^{-E/RT}}{Q_{\max}\, RT} \cdot \frac{1 + (1-\beta)\, a_{RP} (C^*)^{\beta} \mathrm{e}^{-\beta E/RT}}{\left[1 + a_{RP} (C^*)^{\beta} \mathrm{e}^{-\beta E/RT}\right]^2}. \tag{S-176}$$

For $\beta = 1$: reduces to the logistic density (Langmuir). For $\beta < 1$: non-standard asymmetric form with no closed-form σ_E.

S-4.4.10 Koble–Corrigan

Equivalence to Sips. The Koble–Corrigan isotherm $q_e = A_{KC}\, C_e^m \,/\, (1 + B_{KC}\, C_e^m)$ is algebraically identical to Sips with $Q_{\max} = A_{KC}/B_{KC}$ and $K_S = B_{KC}$, same exponent m. The energy distribution and σ_E are therefore those of Section S-4.4.3:

$$\sigma_E = \frac{\pi}{\sqrt{3}} \frac{RT}{m}. \tag{S-177}$$

S-4.4.11 Radke–Prausnitz

Starting isotherm.

$$q_e = \frac{a_{RP}\, r_{RP}\, C_e}{a_{RP} + r_{RP}\, C_e^{p}}, \qquad 0 < p \le 1. \tag{S-178}$$

Distribution. Applying the condensation substitution yields a non-standard distribution without a simple closed form. No analytical σ_E exists; numerical integration is required. For $p \to 0$ the distribution approaches a logarithmic form with diverging width; for $p = 1$ it reduces to the Langmuir (logistic) distribution.

S-4.4.12 Dubinin–Radushkevich / Dubinin–Astakhov

Polanyi potential. D-R and D-A operate in the Polanyi adsorption potential $\varepsilon = RT\ln(1+C_{\mathrm{ref}}/C_e)$ rather than directly in E. Here C_{ref} is a declared reference concentration in the same units as C_e, ensuring that the logarithm is dimensionless. The isotherm is

$$q_e = Q_{\max} \exp\left[-\left(\frac{\varepsilon}{E_a}\right)^{n_{DA}}\right]. \tag{S-179}$$

Derivation of $f(\varepsilon)$. Differentiating with respect to ε:

$$f(\varepsilon) = -\frac{1}{Q_{\max}}\frac{\mathrm{d}q}{\mathrm{d}\varepsilon} = \frac{n_{DA}}{E_a}\left(\frac{\varepsilon}{E_a}\right)^{n_{DA}-1} \exp\left[-\left(\frac{\varepsilon}{E_a}\right)^{n_{DA}}\right]. \tag{S-180}$$

This is the **Weibull** probability density with scale E_a and shape n_{DA}.

Mean adsorption potential.

$$\langle\varepsilon\rangle = \frac{E_a}{n_{DA}}\,\Gamma\!\left(\frac{1}{n_{DA}}\right). \tag{S-181}$$

D-R ($n_{DA} = 2$): $\langle\varepsilon\rangle = E_a\sqrt{\pi}/2$.

Width. The variance of the Weibull distribution with scale E_a and shape n_{DA} is $E_a^2[\Gamma(1+2/n_{DA}) - \Gamma(1+1/n_{DA})^2]$, therefore

$$\sigma_\varepsilon = E_a\sqrt{\Gamma\left(1+\frac{2}{n_{DA}}\right) - \left[\Gamma\left(1+\frac{1}{n_{DA}}\right)\right]^2}. \tag{S-182}$$

D-R ($n_{DA}=2$): $\Gamma(2)=1$, $\Gamma(3/2)=\sqrt{\pi}/2$, so

$$\sigma_\varepsilon = E_a\sqrt{1-\pi/4} \approx 0.4633\,E_a. \tag{S-183}$$

The distribution is right-skewed (longer tail toward high adsorption potentials).

S-4.4.13 Hill

Starting isotherm.

$$q_e = \frac{Q_{\max}\,C_e^{n_H}}{K_D + C_e^{n_H}}. \tag{S-184}$$

Condensation substitution. Setting $C_e = C^* \mathrm{e}^{-E/RT}$ and defining $E_c = RT\ln(K_D^{-1/n_H}\,C^*)$:

$$q(E) = \frac{Q_{\max}}{1+\mathrm{e}^{n_H(E-E_c)/RT}}. \tag{S-185}$$

Differentiation.

$$\frac{\mathrm{d}q}{\mathrm{d}E} = -Q_{\max}\,\frac{(n_H/RT)\,\mathrm{e}^{n_H(E-E_c)/RT}}{\left[1+\mathrm{e}^{n_H(E-E_c)/RT}\right]^2}. \tag{S-186}$$

Distribution.

$$f(E) = \frac{Q_{\max}\,(n_H/RT)\,\mathrm{e}^{n_H(E-E_c)/RT}}{\left[1+\mathrm{e}^{n_H(E-E_c)/RT}\right]^2}. \tag{S-187}$$

Logistic density (times $Q_{\max}$) with location E_c and scale $s = RT/n_H$; identical in form to Sips (Section S-4.4.3) with $m \to n_H$.

Width.

$$E_c = RT\ln(K_D^{-1/n_H}\,C^*), \qquad \sigma_E = \frac{\pi}{\sqrt{3}}\,\frac{RT}{n_H}. \tag{S-188}$$

Physical interpretation.

- $n_H < 1$ (anti-cooperative): $\sigma_E > (\pi/\sqrt{3})\,RT$ — broader than Langmuir.
- $n_H = 1$ (non-cooperative): $\sigma_E = (\pi/\sqrt{3})\,RT \approx 1.81\,RT$.
- $n_H > 1$ (cooperative binding): $\sigma_E < (\pi/\sqrt{3})\,RT$ — narrower distribution; the adsorption transition is sharper.

S-5 Clustering and classification of results

Once the descriptor set $\{Q_{\max},\,\ln K^*_{\mathrm{aff}},\,\sigma_H\}$ has been extracted for many adsorbent–adsorbate systems, each material is represented by a feature vector

$$\mathbf{x}_i = \big(Q_{\max,i},\ \ln K^*_{\mathrm{aff},i},\ \sigma_{H,i}\big)^\top \in \mathbb{R}^d, \qquad d = 3. \tag{S-189}$$

Stacking the N materials produces the descriptor matrix

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N\times d}. \tag{S-190}$$

All clustering and classification methods below operate on this common descriptor representation, so the grouping is driven by physically interpretable quantities rather than by the algebraic form of the original isotherm equation.

Plain-language interpretation

In this three-number space, $Q_{\max}$ answers "how much?", $\ln K^*_{\text{aff}}$ answers "how strongly?", and σ_H answers "how heterogeneous?". These are the quantities the machine-learning methods see; the original isotherm equation is only the route used to estimate them.

S-5.1 Feature standardisation

$$\tilde{x}_{i,\ell} = \frac{x_{i,\ell} - \bar{x}_\ell}{s_\ell}, \qquad s_\ell = \sqrt{\frac{1}{N-1}\sum_i (x_{i,\ell} - \bar{x}_\ell)^2}. \tag{S-191}$$

This transformation centers each descriptor to zero mean and rescales it to unit variance, preventing $Q_{\max}$, $\ln K^*_{\text{aff}}$, or σ_H from dominating the geometry solely because of their numerical units.

S-5.2 Categorical covariates and one-hot encoding

If auxiliary categorical information is available, such as material family, synthesis route, or best-fit isotherm class, it can be appended to the continuous descriptor vector by one-hot encoding. Let $\mathbf{z}_i \in \{0,1\}^q$ denote the binary indicator block for material i. The augmented feature vector is then

$$\mathbf{x}_i^{\text{aug}} = \begin{bmatrix} \tilde{\mathbf{x}}_i \\ \mathbf{z}_i \end{bmatrix} \in \mathbb{R}^{d+q}. \tag{S-192}$$

For a single categorical variable with q levels, the indicator block satisfies $\sum_{j=1}^{q} z_{ij} = 1$. In a linear classifier, the decision surface in the augmented space is therefore a hyperplane in $\mathbb{R}^{d+q}$, exactly as discussed in the main manuscript.

S-5.3 Distance metric

$$d(\mathbf{x}_i, \mathbf{x}_j) = \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|_2 = \sqrt{\sum_{\ell=1}^{d} \left(\frac{x_{i,\ell} - x_{j,\ell}}{s_\ell}\right)^2}. \tag{S-193}$$

When one-hot variables are included, the same Euclidean formula is applied to $\mathbf{x}_i^{\text{aug}}$, optionally after scaling the binary block consistently across all samples.

S-5.4 k-Nearest Neighbours (KNN)

Given a query material $\mathbf{x}_q$, assign it to the class most common among its k nearest neighbours:

$$\hat{y}(\mathbf{x}_q) = \arg\max_c \sum_{i \in \mathcal{N}_k(\mathbf{x}_q)} \mathbf{1}[y_i = c]. \tag{S-194}$$

Here $\mathcal{N}_k(\mathbf{x}_q)$ is the set of the k nearest training points under Eq. (S-193). A practical starting value is $k \approx \sqrt{N}$, rounded to the nearest odd integer for binary classification.

The choice of k should be validated by leave-one-out cross-validation (LOOCV):

$$\text{Acc}_{\text{LOO}}(k) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[\hat{y}_k^{(-i)}(\mathbf{x}_i) = y_i\right], \tag{S-195}$$

where $\hat{y}_k^{(-i)}$ denotes the KNN prediction obtained after removing sample i from the training set. The value of k that maximises Acc_{LOO} is preferred. Distance-weighted voting may be used as a refinement, but the unweighted majority rule of Eq. (S-194) is sufficient for the exploratory use cases presented in the manuscript.

S-5.5 Support Vector Machines (SVM)

For binary labels $y_i \in \{-1, +1\}$, the soft-margin SVM finds the maximum-margin separating hyperplane by solving

$$\min_{\mathbf{w}, b} \frac{1}{2}\|\mathbf{w}\|^2 + C_{\text{reg}} \sum_i \xi_i, \qquad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0. \tag{S-196}$$

The decision function can be written in dual form as

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i y_i \mathcal{K}(\mathbf{x}_i, \mathbf{x}) + b, \qquad \hat{y}(\mathbf{x}) = \operatorname{sign}\big(f(\mathbf{x})\big), \tag{S-197}$$

where only the samples with $\alpha_i > 0$ are support vectors. For a linear kernel, Eq. (S-197) defines a plane in the continuous descriptor space or, when Eq. (S-192) is used, a hyperplane in $\mathbb{R}^{d+q}$.

For non-linearly separable data, use the RBF kernel $\mathcal{K}(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$. Multi-class problems can be handled by one-vs-rest or one-vs-one decompositions, both of which remain compatible with the same descriptor set.

S-5.6 Hierarchical clustering (Ward's linkage)

Agglomerative clustering builds a dendrogram by iteratively merging the two closest clusters. For any cluster C, define the within-cluster sum of squares

$$W(C) = \sum_{i \in C} \|\tilde{\mathbf{x}}_i - \bar{\mathbf{x}}_C\|^2, \qquad \bar{\mathbf{x}}_C = \frac{1}{|C|} \sum_{i \in C} \tilde{\mathbf{x}}_i. \tag{S-198}$$

Ward's linkage chooses the merge that minimises the increase in within-cluster variance:

$$\Delta(A, B) = W(A \cup B) - W(A) - W(B) = \frac{n_A \, n_B}{n_A + n_B} \|\bar{\mathbf{x}}_A - \bar{\mathbf{x}}_B\|^2. \tag{S-199}$$

The dendrogram cut height determines the number of clusters; the "elbow" in the linkage distance curve provides a natural choice, and cluster stability can be checked by re-running the procedure on bootstrap resamples of the descriptor matrix.

S-5.7 Classifier validation and cluster reproducibility

For supervised classifiers, predictive performance should be evaluated by M-fold cross-validation:

$$\mathrm{Acc}_{\mathrm{CV}} = \frac{1}{M}\sum_{m=1}^{M}\frac{1}{|T_m|}\sum_{i\in T_m}\mathbf{1}\left[\hat{y}^{(-m)}(\mathbf{x}_i) = y_i\right], \tag{S-200}$$

where T_m is the test subset in fold m and $\hat{y}^{(-m)}$ is the classifier trained on the remaining folds. If no external class labels exist, the cluster labels obtained from Ward's method can be treated as pseudo-labels and their reproducibility quantified by KNN LOOCV (Eq. (S-195)) or SVM cross-validation (Eq. (S-200)). High reproducibility indicates that the clusters reflect a stable geometric structure in descriptor space rather than an arbitrary cut of the dendrogram.

S-5.8 Low-dimensional visualisation

Principal component analysis (PCA) provides a compact visual summary of the descriptor matrix. For the standardised matrix $\tilde{\mathbf{X}}$, the sample covariance matrix is

$$\mathbf{S} = \frac{1}{N-1}\tilde{\mathbf{X}}^{\top}\tilde{\mathbf{X}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\top}, \tag{S-201}$$

where the columns of $\mathbf{V}$ are the principal axes and $\mathbf{\Lambda}$ contains the eigenvalues. Projection onto the first r principal components is

$$\mathbf{z}_i = \mathbf{V}_r^{\top}\tilde{\mathbf{x}}_i, \tag{S-202}$$

which is convenient for plotting cluster structure in two dimensions without changing the underlying classifier or clustering definitions.

S-5.9 Workflow summary for clustering

1. For each material, compute $\{Q_{\max},\, K^{*}_{\mathrm{aff}},\, \sigma_H\}$.

2. Standardise all features (Eq. S-191).

3. If categorical metadata are used, append them through one-hot encoding (Eq. S-192).

4. Compute the pairwise distance matrix (Eq. S-193).

5. Apply hierarchical clustering (Ward's linkage) and cut the dendrogram.

6. Validate clusters using KNN leave-one-out cross-validation (Eq. S-195) or SVM with cross-validation (Eq. S-200).

7. Visualise clusters in 2D projections (Q_{max}, $\ln K^*_{\text{aff}}$) or via PCA (Eq. S-202).

Classification versus clustering

KNN and SVM are **classification** tools: they need labels such as high/medium/low performance or material family. Ward's linkage is **clustering**: it searches for groups when labels are not known. Use the word "classification" for KNN/SVM and "clustering" for Ward.

S-6 Complete reporting protocol

For each adsorbent–adsorbate system, the following information should be reported.

Minimum message for readers

A complete result should let another group reproduce the fit and compare the material. Report the raw conditions, the selected model, the fit quality, and the four final descriptors $\{Q_{\text{max}}, K^*_{\text{aff}}, E_{\text{ads}}, \sigma_H\}$.

S-6.1 Minimum reporting set

Table S-3: Minimum reporting set for adsorption isotherm analysis.

Category	Quantity	Units	Source
Experimental	T, pH, ionic strength	K, —, M	Measured
	C° (declared)	$\mathrm{mg\,L^{-1}}$ or $\mathrm{mol\,L^{-1}}$	Convention
	n (data points)	—	Measured
Model fit	Best model name	—	Step 4
	Fitted parameters $\boldsymbol{\theta}$	model-dependent	Step 2
	R^2, SSE, ARED, AIC$_c$	—	Step 3
Universal	$Q_{\max}$	$\mathrm{mg\,g^{-1}}$	Step 5
	K^*_{aff}	dimensionless	Step 5
	E_{ads}	$\mathrm{kJ\,mol^{-1}}$	Step 5
	σ_H	$\mathrm{kJ\,mol^{-1}}$	Step 5
Comparison	OVL (if comparing models)	$[0, 1]$	Step 6
	Cluster label	—	Step 7

S-6.2 Extracting descriptors from each model

Name used in the final report

The final manuscript uses E_{ads}. If an intermediate derivation uses E_{aff}, read it as the same positive affinity-energy descriptor:

$$E_{\mathrm{ads}} = E_{\mathrm{aff}} = RT \ln K^*_{\mathrm{aff}}.$$

Table S-4: Extraction of operational reporting descriptors from native isotherm parameters. All expressions assume C° is the declared standard-state concentration.

Model	$\boldsymbol{K_{\text{aff}}}$ $\boldsymbol{([C]^{-1})}$	$\boldsymbol{E_{\text{ads}}/(RT)}$	$\boldsymbol{\sigma_H/(RT)}$
Langmuir	K_L	$\ln(K_L C^\circ)$	0
Double Langmuir	$K_L^{(\text{tot})} = \dfrac{Q_1K_1 + Q_2K_2}{Q_1 + Q_2}$	$\ln(K_L^{(\text{tot})} C^\circ)$	$\dfrac{\sqrt{Q_1Q_2}}{Q_1+Q_2}\dfrac{\lvert\Delta E_{12}\rvert}{RT}$
Freundlich	$\dfrac{1}{n}K_F C_{\text{ref}}^{1/n-1}$ [a]	$\ln\left[\dfrac{1}{n}K_F C_{\text{ref}}^{1/n-1}C^\circ\right]$ [a]	∞
Sips	$K_S^{1/m}$	$\dfrac{1}{m}\ln(K_S(C^\circ)^m)$	$\dfrac{\pi}{\sqrt{3}}\sqrt{\dfrac{1}{m^2}-1}$
Tóth	b	$\ln(b\,C^\circ)$	numerical
Jovanovic	b	$\ln(b\,C^\circ)$	$\frac{\pi}{\sqrt{6}} \approx 1.28$
Temkin	a_T	$\ln(a_T C^\circ)$	Eq. S-168 [b]
UNILAN	$\sqrt{b_{\max}b_{\min}}$	$\frac{1}{2}\ln(b_{\max}b_{\min}(C^\circ)^2)$	$\frac{\ln(b_{\max}/b_{\min})}{2\sqrt{3}}$
Redlich–Peterson	a_{RP} [c]	$\ln(a_{RP}C^\circ)$ [c]	—
Koble–Corrigan	B_{KC}^n	$n\ln(B_{KC}C^\circ)$	$\dfrac{\pi}{\sqrt{3}}\sqrt{n^2-1}$
Radke–Prausnitz	r_{RP} [d]	$\ln(r_{RP}C^\circ)$ [d]	—
Hill	K_D^{-1/n_H}	$-\frac{1}{n_H}\ln K_D + \ln C^\circ$	$\dfrac{\pi}{\sqrt{3}}\sqrt{\max\{1/n_H^2-1,0\}}$
D-R/D-A	$1/C_{1/2}$	$\ln(C^\circ/C_{1/2})$	Eq. S-182

[a] Apparent, reference-dependent quantity; no intrinsic $Q_{\max}$ or universal affinity exists for Freundlich; any Freundlich affinity is local and reference-dependent.
[b] Requires a declared effective Temkin window or $Q_{\max}^{\text{eff}}$; not intrinsic to the two-parameter fit.
[c] Henry-based cross-model comparison is valid only in the Langmuir limit $\beta = 1$; otherwise Redlich–Peterson has no finite $Q_{\max}$.
[d] Effective low-concentration affinity; Radke–Prausnitz has no finite $Q_{\max}$.

For Sips, Koble–Corrigan, and Hill, the table reports the intrinsic descriptor σ_H, not the apparent logistic width σ_E. For other models without a Langmuir thermal-kernel correction, the reported width is the physically relevant model width or is left undefined when no reliable closed form exists.

S-6.3 Key message

Final reporting fingerprint

Regardless of which isotherm model best fits the data, the final report should reduce the result to the same descriptor set:

$$\{Q_{\max},\, K^*_{\mathrm{aff}},\, E_{\mathrm{ads}},\, \sigma_H\}. \tag{S-203}$$

Here $Q_{\max}$ says how much can be adsorbed, K^*_{aff} and E_{ads} place affinity on a declared standard-state scale, and σ_H reports intrinsic heterogeneity. This four-number fingerprint is the bridge between model fitting, cross-study comparison, uncertainty analysis, and descriptor-based classification.

S-7 Companion notebooks and usage

The repository accompanying this manuscript is intended to let the user fit isotherm data, evaluate competing models, extract descriptors, and build classification datasets without rederiving the equations in this SI.

S-7.1 Repository contents

- `01_isotherm_fitting.ipynb`: fits the candidate-model library to raw adsorption data with columns `Ce` and `qe`, then separates converged diagnostic fits from physically admissible reporting models.

- `02_error_metrics_model_selection.ipynb`: computes error metrics, information criteria, F-tests, and the reporting table of descriptors.

- `03_statistical_validation.ipynb`: performs bootstrap, noise-sensitivity, and descriptor-uncertainty propagation.

- `04_classification_clustering.ipynb`: applies KNN, SVM, and Ward clustering to descriptor tables.

- `app_isotherm_fitting.py`: Streamlit app for interactive fitting and export of both fit metrics and descriptor tables.

- `utils_isotherms.py`: shared implementation of model equations, error metrics, descriptor extraction, and distribution utilities.

S-7.2 Notebook output blocks used in this SI

The notebooks are part of the scientific record, but the SI should not display raw notebook exports or raw LaTeX setup code. The reader needs to see what was run, what came out, and where the reproducible file is stored. Therefore, the notebook material in this SI is represented as cleaned notebook-style blocks: short input cells, short output cells, exported figures, exported tables, and a link back to the executable notebook in the repository.

> **Rule for notebook material in the SI**
>
> A notebook block must answer three questions: What file or cell was run? What result did it produce? Which manuscript concept does it support? Long imports, package installation commands, plotting boilerplate, and raw LaTeX commands belong in the repository or build notes, not in the reader-facing SI.

Notebook block 1: minimal fitting cell. This is the type of short code block that belongs in the SI. It shows the scientific operation without reproducing the whole notebook.

Notebook input cell: fitting and descriptor extraction

```
In [1]:  import pandas as pd

In [2]:  from utils_isotherms import fit_all_models, extract_descriptors

In [3]:  data = pd.read_csv("sample_isotherm_data.csv")

In [4]:  fits = fit_all_models(data["Ce"], data["qe"])

In [5]:  descriptors = extract_descriptors(fits, temperature_K=298.15)
```

Notebook output cell: reported descriptor table

```
Out [5]:  columns = model, Qmax, K_aff_star, E_ads, sigma_H, AICc, BIC

Out [5]:  sigma_H is the reported heterogeneity width; legacy sigma_E is not
used as the final descriptor.
```

Notebook block 2: figure export. Figures should appear in the SI as figures, not as screenshots of the notebook interface. The notebook cell should save the figure with a stable filename; the SI should show the exported figure and explain what it demonstrates.

Notebook input cell: export a fit overlay

```
In [6]:  ax = plot_fit_overlay(data, fits["Sips"])

In [7]:  ax.figure.savefig("SI_notebook01_fit_overlay.pdf",
bbox_inches="tight")
```

Notebook output artifact

```
Out [7]:  SI_notebook01_fit_overlay.pdf
```

The corresponding SI figure should use the exported file and caption the scientific point: agreement between measured adsorption data and the selected model, residual structure, or comparison among candidate isotherms.

Notebook block 3: table export. Tables should be exported from the notebook after rounding, unit normalization, and descriptor naming have been checked.

Notebook input cell: export a model-selection table

```
In [8]:  report = build_model_selection_table(fits)

In [9]:  report.to_csv("SI_notebook02_model_selection.csv", index=False)

In [10]:  report.to_latex("SI_notebook02_model_selection_table.tex",
index=False)
```

Notebook output artifact

```
Out [10]:  SI_notebook02_model_selection_table.tex
```

The SI table should contain the final reported columns only: model name, fitted capacity, standardized affinity, adsorption-energy descriptor, intrinsic heterogeneity width σ_H, and model-selection statistics.

Notebook block 4: statistical validation. Bootstrap and noise-sensitivity results should be reduced to intervals and short diagnostic plots. The SI should not include the entire simulation log.

Notebook input cell: bootstrap summary

```
In [11]:  boot = bootstrap_fit(data, model="Sips", n_boot=1000)

In [12]:  boot_summary = summarize_descriptor_intervals(boot)

In [13]:  boot_summary[["E_ads", "sigma_H"]]
```

Notebook output cell: uncertainty statement

```
Out [13]:  E_ads median and 95 percent interval

Out [13]:  sigma_H median and 95 percent interval
```

The text should report whether the descriptor ranking is stable under bootstrap resampling, not merely whether a flexible model can fit the data.

Notebook block 5: classification and clustering. Classification notebooks should show the descriptor columns and the target labels before showing accuracy, confusion matrices, PCA maps, or clusters.

Notebook input cell: descriptor matrix for classification

```
In [14]:  X = descriptors[["Qmax", "K_aff_star", "E_ads", "sigma_H"]]
In [15]:  y = descriptors["group"]
In [16]:  scores = cross_validate_svm_knn(X, y)
```
677

Notebook output cell: classification summary

```
Out [16]:  cross-validated KNN and SVM scores
Out [16]:  confusion matrix and PCA descriptor map exported as figure files
```

The SI should interpret whether the separation is driven by capacity, affinity, adsorption energy, or intrinsic heterogeneity, rather than treating the classifier as a black box.
678

What not to include

Do not include raw `nbconvert` pages, long import cells, package installation logs, hidden execution state, or LaTeX preamble fragments. Those items are useful for building the supplement but they are not scientific content for the reader. The clean SI artifact is the notebook-style block plus the exported figure or table.
679

Codex–Overleaf working convention

Run and maintain the analysis in Jupyter. Export figures as stable PDF or PNG files and export summary tables as CSV or cleaned table files. In the SI, include only short notebook-style blocks that explain the reproducible action and the output artifact. Keep the complete notebooks in the repository as the executable supplement.
680

681 ## S-7.3 Notebook-to-SI traceability

Code-to-text rule

Every notebook operation is either a direct implementation of an equation in this SI or a diagnostic visualisation of those same quantities. The notebooks should not introduce new physical descriptors beyond $\{Q_{\max}, K^*_{\mathrm{aff}}, E_{\mathrm{ads}}, \sigma_H\}$.
682

Table S-5: Traceability between active notebook operations and the SI sections where the corresponding concepts are defined.

Notebook or script	Implemented concept	Where it is explained
`01_isotherm_fitting.ipynb`	Non-linear fitting of the candidate isotherm equations; overlay plots of experimental and fitted curves; extraction of $\{Q_{\max}, K^*_{\text{aff}}, E_{\text{ads}}, \sigma_H\}$.	Model equations in Section S-1; descriptor extraction in Section S-6.1.
`02_error_metrics_model_selection.ipynb`	SSE, R^2, adjusted R^2, EABS, RESID, ARED, MPSED, HYBRID, AIC, AIC$_c$, BIC, Akaike weights, F-tests, residual plots, parity plots, metric bar/radar summaries, heatmaps, and covariance confidence bands.	Error metrics and confidence bands in Section S-2; model selection in Section S-3.
`03_statistical_validation.ipynb`	Residual bootstrap, confidence intervals, synthetic-noise sensitivity, and Monte Carlo propagation from fitted parameters to E_{ads} and σ_H.	Bootstrap and noise sensitivity in Section S-2; final descriptors in Section S-6.1.
`04_classification_clustering.ipynb`	Standardisation, legacy σ_E-to-σ_H column renaming, one-hot encoding of categorical metadata, KNN, SVM, Ward clustering, cross-validation, confusion matrices, and PCA visualisation.	Clustering and classification in Section S-5.
`utils_isotherms.py` `app_isotherm_fitting.py`	Shared implementation of the model equations, including Sips with $K_S C_e^m$, Dubinin with $\varepsilon = RT \ln(1 + C_{\text{ref}}/C_e)$, descriptor extraction, and export of metric and descriptor tables.	Sips and Dubinin conventions in Section S-1; width convention in Section S-4; reporting protocol in Section S-6.1.

Diagnostic graphics such as radar charts, heatmaps, parity plots, residual plots, confusion matrices, and PCA maps are interpretive aids. They do not replace the reporting table; they visualise the residual metrics, uncertainty estimates, or descriptor-space classifications defined above.

S-7.4 Recommended usage workflow

1. Prepare a raw isotherm file in `.csv` or Excel format with columns `Ce` and `qe`.
2. Fit all candidate models using `01_isotherm_fitting.ipynb` or the Streamlit app.
3. Use `02_error_metrics_model_selection.ipynb` to rank models with SSE, AIC_c, BIC, Akaike weights, F-tests, and the descriptor set $\{Q_{\mathrm{max}}, K^*_{\mathrm{aff}}, E_{\mathrm{ads}}, \sigma_H\}$.
4. Use `03_statistical_validation.ipynb` to attach bootstrap intervals and sensitivity tests to the fitted parameters and derived descriptors.
5. When several materials are available, assemble a descriptor table and analyze it with `04_classification_clustering.ipynb`.

S-7.5 Input formats

For raw fitting, the minimum required columns are `Ce` and `qe`. For classification and clustering, the input table should contain at least two numeric descriptor columns and one target column named `group`; optional categorical columns such as material family or best-fit isotherm type are one-hot encoded automatically in the notebook.

S-7.6 Sample files

The repository includes `sample_isotherm_data.csv` for raw fitting and three descriptor tables for classification: `sample_descriptors_basic.csv`, `sample_descriptors_isotherm.csv`, and `sample_descriptors_full.csv`. Older notebook variants stored in `backup_notebooks/` are archival only and are not part of the active workflow.

A Notebook 01: fitting one isotherm

File: `01_isotherm_fitting.ipynb`.

Purpose of this notebook

Use this notebook when the starting point is one experimental isotherm: a table of equilibrium concentration C_e and adsorption capacity q_e. Its job is to fit the candidate isotherm equations, compare the fitted curves visually, and extract the universal descriptors from each converged model.

Input file

Place a `.csv`, `.xlsx`, or `.xls` file in the notebook folder. The file must contain at least two columns:

- `Ce`: equilibrium concentration, in one declared unit such as $\mathrm{mg\,L^{-1}}$ or $\mathrm{mol\,L^{-1}}$;
- `qe`: adsorbed amount, in one declared unit such as $\mathrm{mg\,g^{-1}}$.

Column names are treated case-insensitively. The notebook removes missing values and keeps only $C_e > 0$ and $q_e \geq 0$.

How to run it

1. Set `DATA_FILE` to the experimental file name.
2. Run the import and data-loading cells.
3. Run the fitting cell. The notebook fits all implemented models by non-linear least squares; the equations are those listed in Section S-1.
4. Inspect the fit plot and residual plot before accepting the numerical ranking. See Figure S-2
5. Read the descriptor table. The reportable descriptors are $\{Q_{\max}, K^*_{\mathrm{aff}}, E_{\mathrm{ads}}, \sigma_H\}$.

What to look at first

For a materials reader, the first-pass answer is: which model fits without systematic residuals, what $Q_{\max}$ is obtained, whether E_{ads} is comparable to related systems, and whether σ_H is close to zero or clearly heterogeneous.

Common mistake

Do not choose the model only because it gives the highest R^2. A more flexible model can fit noise. Use this notebook together with `02_error_metrics_model_selection.ipynb`.

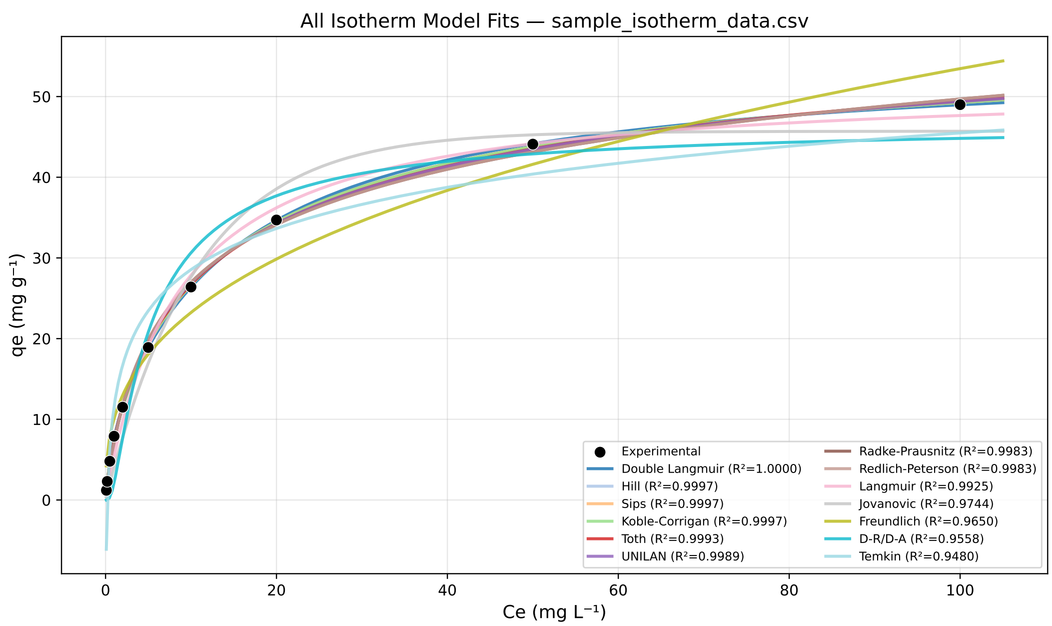

Figure S-2: Overlay of all converged isotherm model fits generated directly from `01_isotherm_fitting.ipynb`. Experimental data are shown together with the fitted model curves, ordered by increasing SSE in the legend.

B Notebook 02: error metrics and model selection

File: `02_error_metrics_model_selection.ipynb`.

Purpose of this notebook

Use this notebook to decide which fitted isotherm should be reported. It computes absolute and relative error metrics, information criteria, Akaike weights, nested-model tests, and diagnostic plots.

Input file

Use the same raw isotherm file as in Appendix A. The same C_e and q_e requirements apply. The notebook refits the models internally so that all metrics are computed from the same dataset and the same candidate set.

How to run it

1. Set `DATA_FILE`.
2. Run all fitting and metric cells.
3. Rank models using AIC_c as the primary statistical criterion.
4. Check whether models within $\Delta\mathrm{AIC}_c < 2$ are physically distinguishable by their descriptor values and energy distributions.
5. Use F-tests only for nested model pairs, such as Langmuir versus Sips, and only after confirming that both models converged sensibly.

Output plots

The notebook produces fit overlays, residual plots, grouped metric bars, AIC_c/Akaike-weight summaries, radar charts, parity plots, heatmaps, and covariance-based confidence bands. These figures are diagnostic aids:

- residual plots reveal systematic model failure; See Figure S-3

- parity plots show whether predictions follow the experimental values; See Figure S-4
- heatmaps and radar charts summarize many metrics compactly; See Figure S-5 and Figure S-6
- confidence bands show how much uncertainty is implied by the local covariance matrix. See Figure S-7

Decision rule

Report the simplest physically meaningful model among the statistically supported candidates. A lower AIC_c is useful only if the resulting parameters and descriptors remain physically interpretable.

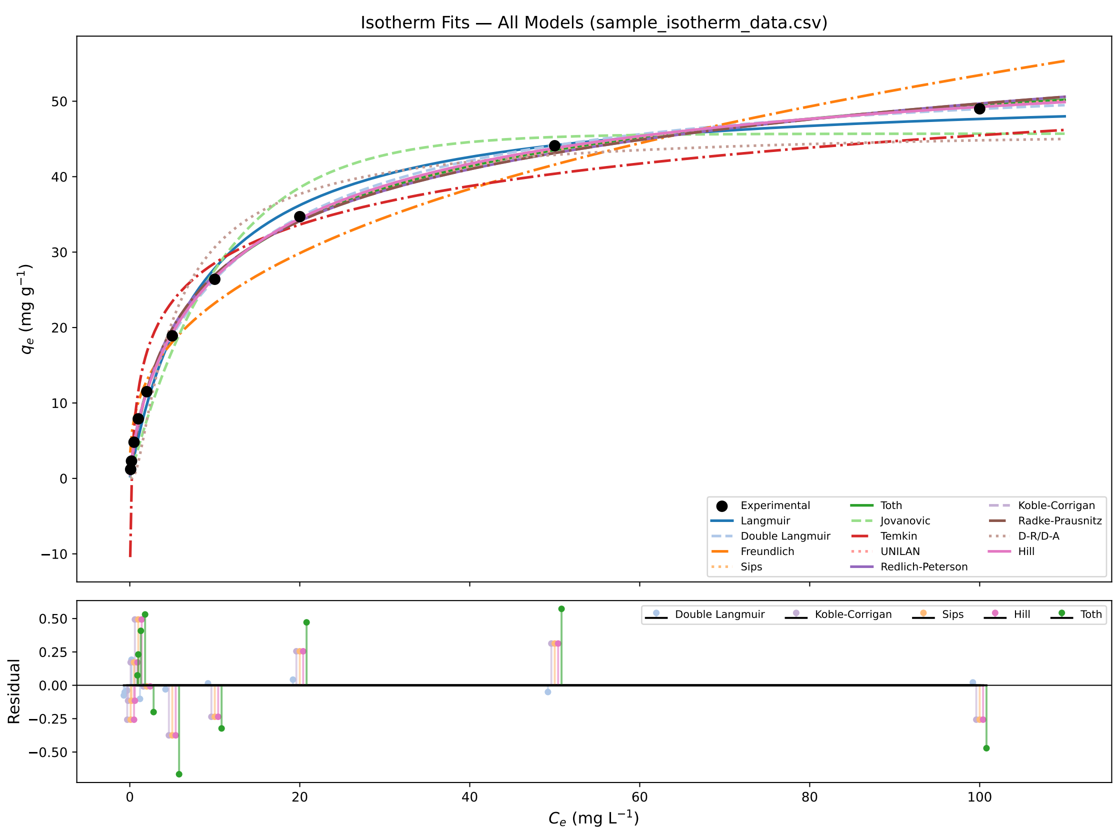

Figure S-3: All converged isotherm model fits and residual analysis generated directly from `01_isotherm_fitting.ipynb`. The upper panel shows the experimental adsorption data together with the fitted model curves. The lower panel shows residuals for the five best-ranked models according to AICc.

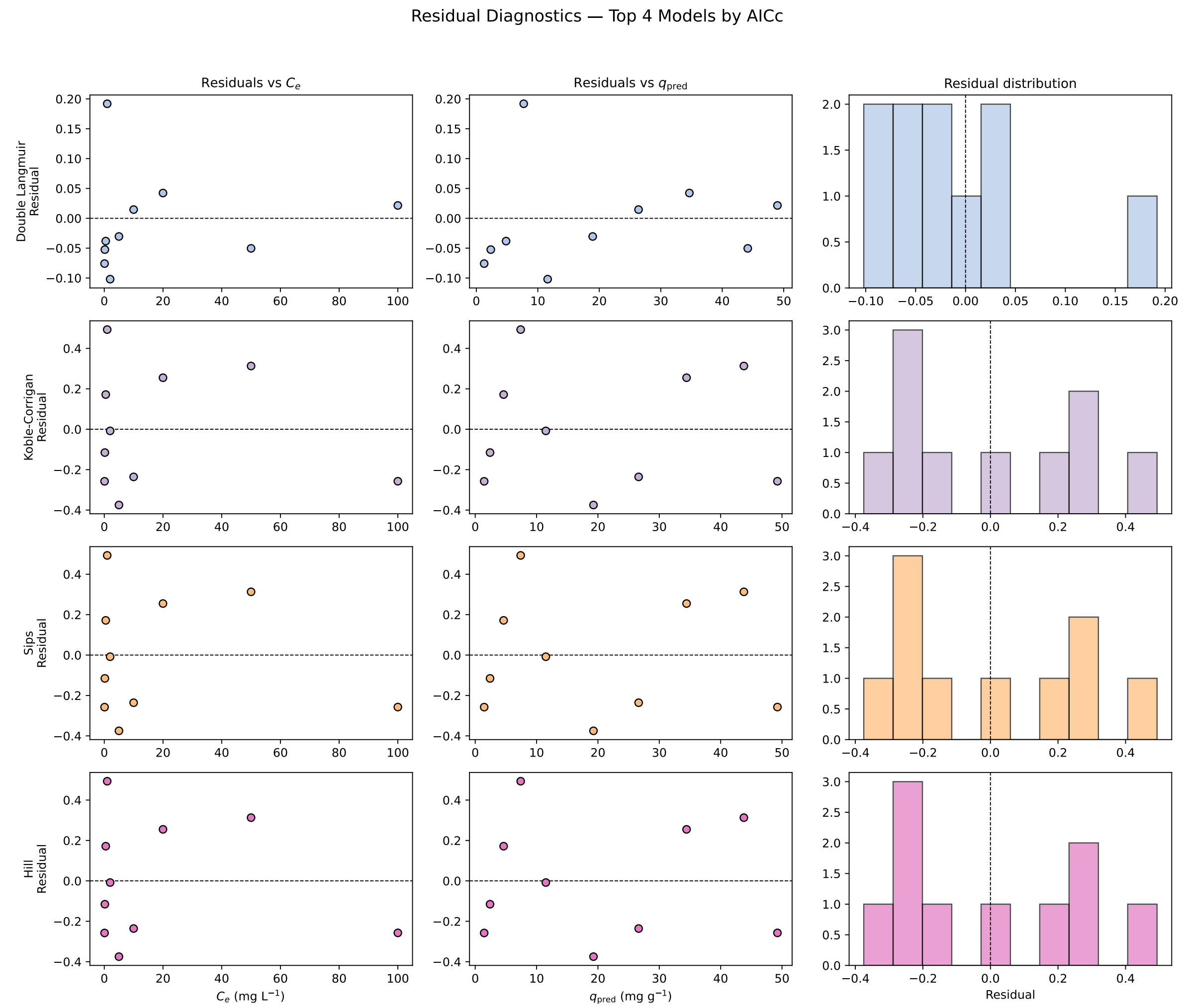

Figure S-4: Residual diagnostics for the four best-ranked isotherm models according to AICc. Each row corresponds to one model. The columns show residuals as a function of equilibrium concentration, residuals as a function of predicted adsorption capacity, and the residual distribution.

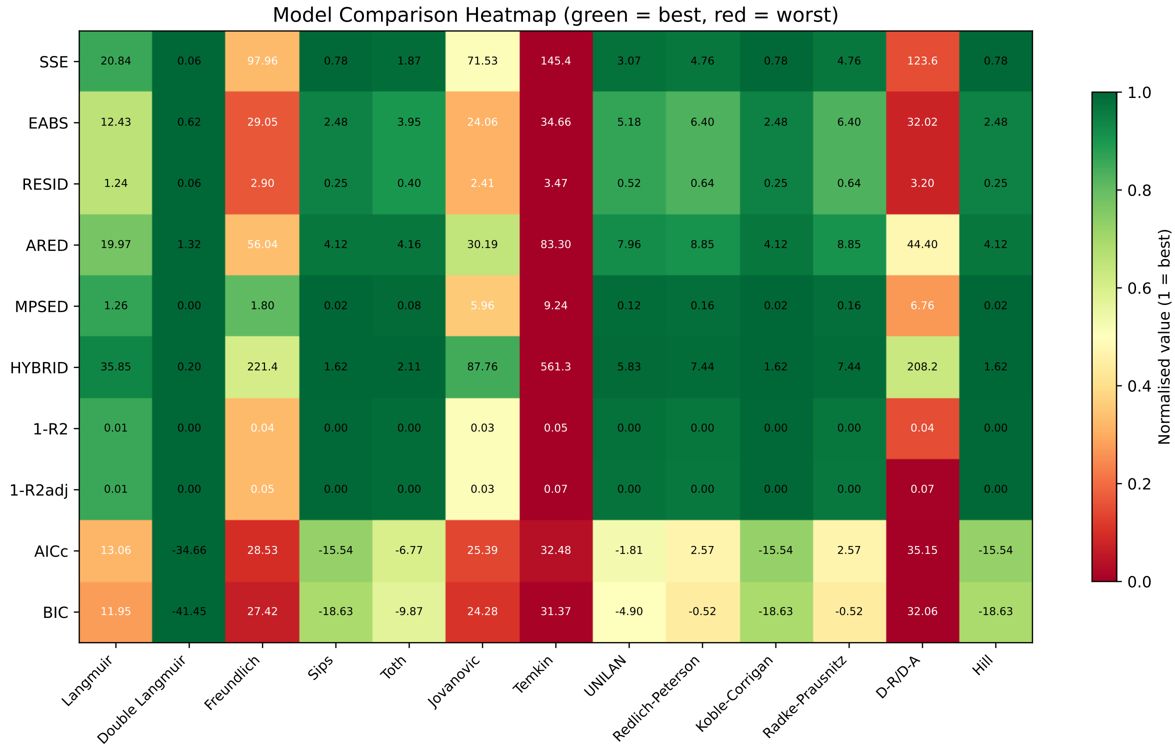

Figure S-5: Model comparison heatmap generated directly from `01_isotherm_fitting.ipynb`. Each row corresponds to a fitting metric or information criterion, normalized independently so that green indicates the best-performing model and red indicates the worst-performing model for that metric.

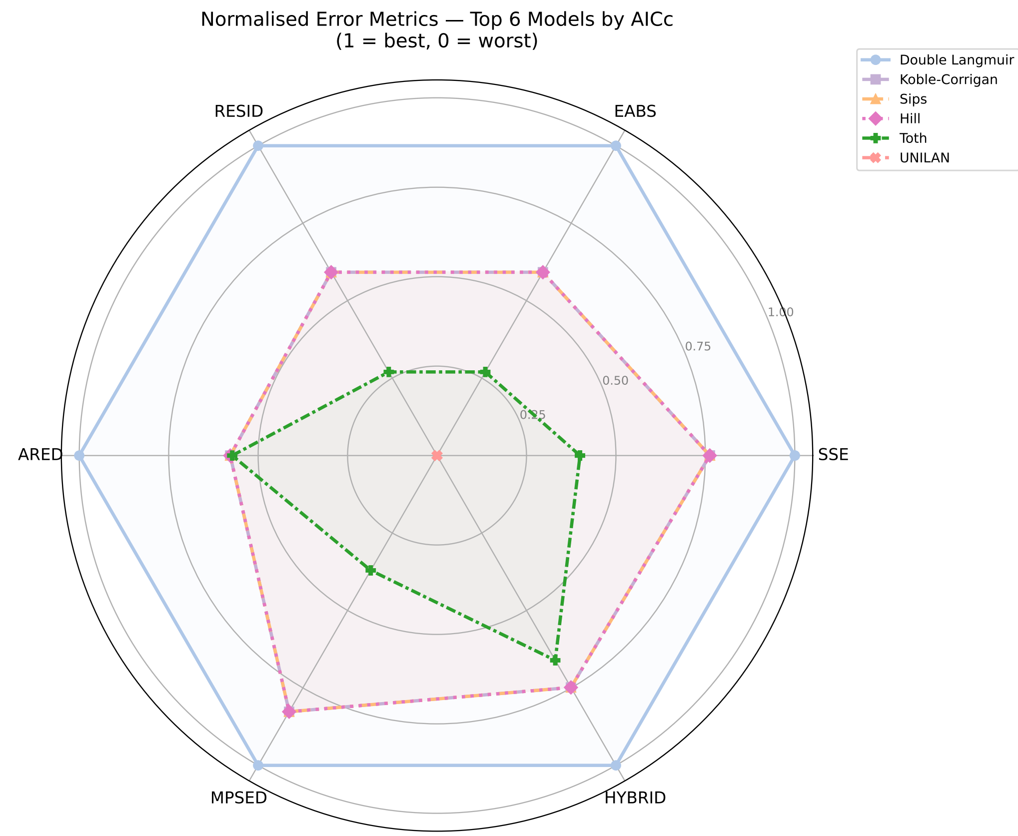

Figure S-6: Radar chart comparing the six best-ranked isotherm models according to AICc. The metrics were independently normalized so that 1 indicates the best-performing model and 0 the worst-performing model for each error criterion.

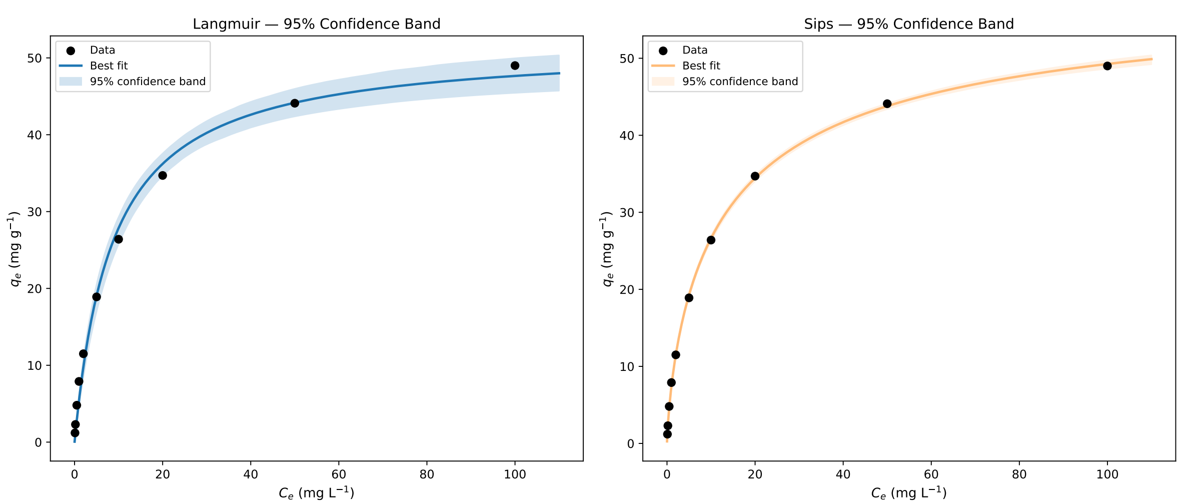

Figure S-7: Confidence bands for Langmuir and Sips isotherm fits generated directly from `01_isotherm_fitting.ipynb`. The solid lines show the best-fit curves, while the shaded regions indicate the 95% confidence bands estimated by Monte Carlo sampling from the fitted parameter covariance matrices.

C Notebook 03: statistical validation

File: `03_statistical_validation.ipynb`.

Purpose of this notebook

Use this notebook after a model has been selected. It asks whether the fitted parameters and descriptors are stable with respect to residual resampling and added measurement noise.

What it tests

The notebook focuses on the Sips model as a worked example because Sips links directly to the homogeneous Langmuir limit through $m \to 1$ and $\sigma_H \to 0$. The same validation logic applies to other fitted models:

- residual bootstrap gives empirical confidence intervals;
- synthetic-noise tests show whether the fitted parameters drift when the response is perturbed;
- Monte Carlo propagation converts parameter uncertainty into uncertainty in E_{ads} and σ_H.

How to use the result

1. Run the baseline fit.
2. Run the bootstrap cell and check the confidence intervals for $Q_{\max}$, K_S, and m.
3. Run the noise-sensitivity cell and check whether the parameter shift is small compared with the bootstrap spread.
4. Report descriptor uncertainty for E_{ads} and σ_H, not only fitted-parameter uncertainty.

Interpretation warning

A narrow confidence interval on a fitted parameter does not automatically mean that the physical descriptor is robust. Always inspect propagated uncertainty in E_{ads} and σ_H.

D Notebook 04: classification and clustering

File: `04_classification_clustering.ipynb`.

Purpose of this notebook

Use this notebook when several materials, adsorbates, or operating conditions have already been converted into descriptor rows. It does not fit isotherms. It classifies or clusters descriptor tables.

Input file

The input table should contain:

- at least two numeric descriptor columns, preferably selected from Q_{max}, E_{ads}, $\ln K^*_{\mathrm{aff}}$, and σ_H;
- one target column named `group` for supervised classification;
- optional categorical columns such as `material_type` or `isotherm_type`.

If an older file contains a column named `sigma_E`, the notebook renames it to `sigma_H` for classification. This is only a backward-compatibility step; new tables should use `sigma_H`.

How to run it

1. Set `DATA_FILE` to one of the descriptor tables.
2. Confirm that `TARGET_COL = "group"` or change it to the desired label column.

3. Run the preprocessing cells. Numeric descriptors are standardised; categorical columns are one-hot encoded.
4. Run KNN for local-neighbour classification.
5. Run SVM for margin-based classification.
6. Run Ward clustering when no trusted labels are available or when the goal is exploratory grouping.
7. Use PCA only as a visual summary of high-dimensional descriptor space, not as a new physical model.

Classification versus clustering

KNN and SVM require labels such as high/medium/low performance or material family. Ward clustering does not require labels; it asks whether the descriptor space contains natural groups.

Common mistake

Do not mix raw fitted parameters with universal descriptors in the same classification table. Use descriptors with consistent units and the same declared C°. Otherwise, the classifier may separate unit conventions rather than materials.